



Medisynix

An AI-powered medical assistant that connects patients, doctors and administrators in one place.

[Complete user journey](#)

[Patient → Doctor → Admin](#)

[Market research · Pros & cons · Roadmap](#)

Final-year project demonstration guide · October 2026

Today's plan (about 15 minutes)

1

The idea · 2 min

Why we built Medisynix, who it helps and how it works under the hood.

2

Patient · 5 min

Sign in, check health, ask the AI, book a doctor, upload a report.

3

Doctor · 3 min

See the day's schedule, spot risky patients, complete a visit.

4

Admin · 2 min

Manage users, check system health, export reports.

5

Market & future · 2 min

What else exists, why we are different, what we do next.

One-sentence pitch

"Medisynix helps people understand their health in plain language and reach the right doctor, while giving doctors and administrators the tools to run the care."

How to read this deck: every journey slide has **Do this** (what to click), **Say** (what to tell the audience) and **Behind the scenes** (what is really happening, for questions).

Before you start: set-up in 3 minutes

1. START THE APP (PRODUCTION MODE IS FASTEST)

```
cd "E:\Personal Projects\AI-powered-medical-assistant"
npm run demo:reset      # fresh demo data, dated for today
npm run build           # once, about 1 minute
npm start               # opens on http://localhost:3000
```

2. OPEN CHROME AT 100% ZOOM, FULL SCREEN (F11)

- Internet is needed for the AI Doctor chat and fonts. MongoDB is **not** needed: the app falls back to local files.
- Open three tabs, one per role. The app remembers one login per browser address, so use **localhost:3000** for the patient, **127.0.0.1:3000** for the doctor and a private (incognito) window for the admin.
- Do not click the Google / GitHub buttons: they need OAuth keys that are not configured. Use email and password.

DEMO ACCOUNTS

Role	Email	Password
Patient	ayesha.khan@demo.medisynix.test	Demo@1234
Doctor	dr.sarah@medisynix.com	doctor123
Admin	admin@medisynix.com	admin123

On the login page choose the matching tab (Patient / Doctor / Admin) first. A mismatch shows "Invalid credentials".

Tip: the full story for each role is on the next slides. If anything goes wrong, the screenshots in this deck show exactly what you should see.

Why we built it: healthcare is hard to reach and hard to understand

256 M

people in Pakistan

~319,000

registered doctors for the whole country

< 1% of GDP

public spending on health

≈ 53%

of health spending paid out
of pocket

117 M

internet users — already
online

Hard to reach

Long queues, travel, and not knowing which doctor or hospital to choose. Many people only go when it is already serious.

Hard to understand

Lab reports and medical words are confusing. People search random websites or ask relatives, and often get wrong or scary answers.

Hard to manage

Records live on paper or in WhatsApp photos. Doctors lack a quick, organised view of a patient's vitals, medicines and history.

Sources: DataReportal Digital 2026 (internet users, population), Pakistan Economic Survey 2024-25 (doctors, health spending), Pakistan National Health Accounts 2019-20 (out-of-pocket share). See the sources slide.

Our solution: one platform, three roles, an AI helper

Patient

- Ask the AI Doctor in plain language
- Check symptoms with safety warnings
- Upload a report and get a simple explanation
- Find a doctor and book a time slot
- Keep vitals, records and medicines in one place

Doctor

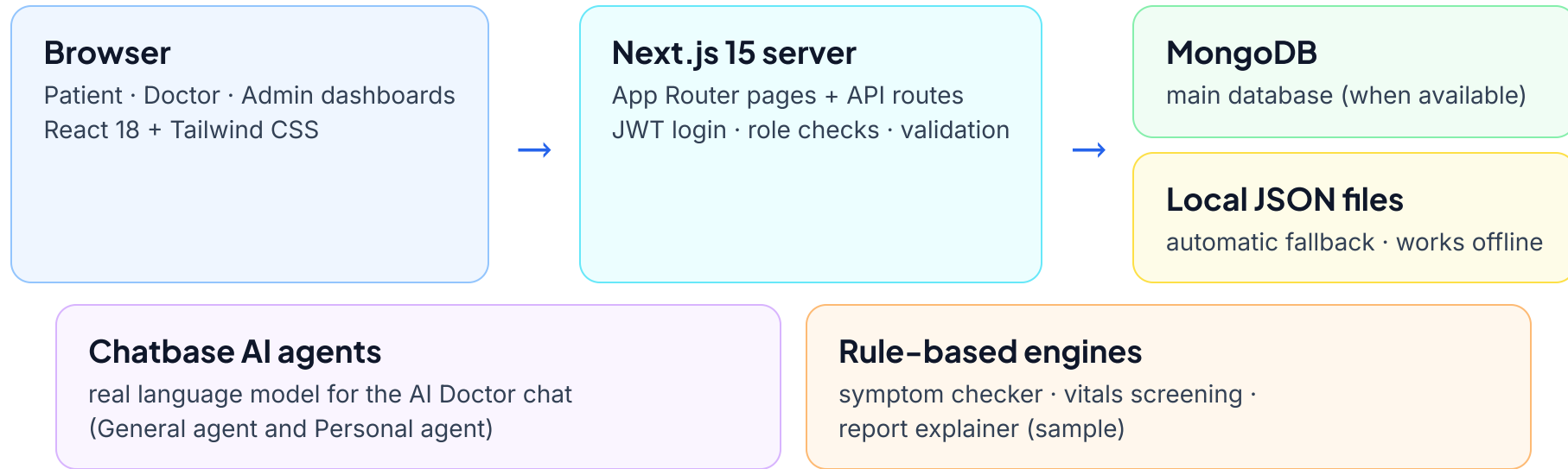
- See today's schedule at a glance
- Vitals screening flags who needs attention
- Complete visits with diagnosis and notes
- Send reports straight to the patient
- Manage public profile, fees and time slots

Administrator

- Create, edit and remove accounts
- Platform health, alerts and security score
- Analytics on users and appointments
- Download reports (CSV / JSON)
- Protect accounts with guard-rails

Design principles: plain language · always show a "not a medical diagnosis" reminder · never invent data · a human doctor stays in charge.

How it is built



Security

Passwords hashed with bcrypt. Signed tokens (7-day expiry). Every API checks the user's role; one patient can never open another patient's data.

Resilience

If the database is down, the app keeps working from local files, and the admin Security page says so honestly.

Tech list

Next.js 15 · React 18 · Tailwind 3 · Recharts · MongoDB · bcryptjs · jsonwebtoken · react-hot-toast · Chatbase.

What is real and what is a demo (be upfront — it builds trust)

Feature	How it works today	Status
AI Doctor chat (General & Personal)	Real language model through Chatbase, grounded on Aga Khan University Hospital and AI Shifa Hospital information	Real AI
Symptom checker	Rule-based: matches your symptoms against 10 common conditions, shows % match, advice and emergency red flags	Rule-based
Vitals screening (doctor view, patient dashboard)	Rules on blood pressure, heart rate, glucose and BMI using standard reference ranges	Rule-based
Report explainer (upload report)	Shows a sample plain-language explanation for the chosen test type with confidence and reasoning; the uploaded file is not yet read by a model	Prototype
Accounts, booking, records, medicines, consultations	Fully working: validated, role-protected, stored in database or local files	Working
Payments, video calls, live chat with doctors	Not built yet — fees are shown for information only	Roadmap

If asked "is the AI real?" — "The conversational assistant is a real language model. The symptom checker and vitals screening are transparent rules we can explain line by line. The report explainer is a prototype that shows the experience we will connect to a medical model next."

The big picture: how the three roles connect

Patient

1. Registers, fills the health profile
2. Finds a doctor and books a slot
3. Uploads a report, asks the AI

Doctor

1. Sees the booking and the patient's vitals
2. Completes the visit with diagnosis & notes
3. Uploads a report for the patient

Admin

1. Creates and manages accounts
2. Watches platform health and alerts
3. Exports reports



What the audience should notice: everything the patient does shows up for the doctor, and everything the doctor does shows up for the patient. The administrator keeps the whole system safe. One shared data model — nothing is typed in twice.

PATIENT

Ayesha Khan, 38

Has high blood pressure and a penicillin allergy. Wants quick, understandable answers and an easy way to see a cardiologist.

What this person wants: Understand my health, ask questions safely, and book and track my care without confusion.

1 Sign in

2 Dashboard

3 Health profile

4 Find a doctor

5 Book a visit

6 AI Doctor

7 Symptom checker

8 Upload report

9 Records & medicines

10 Visits & insights

Welcome and sign in

Show the public website, then log in as a patient.

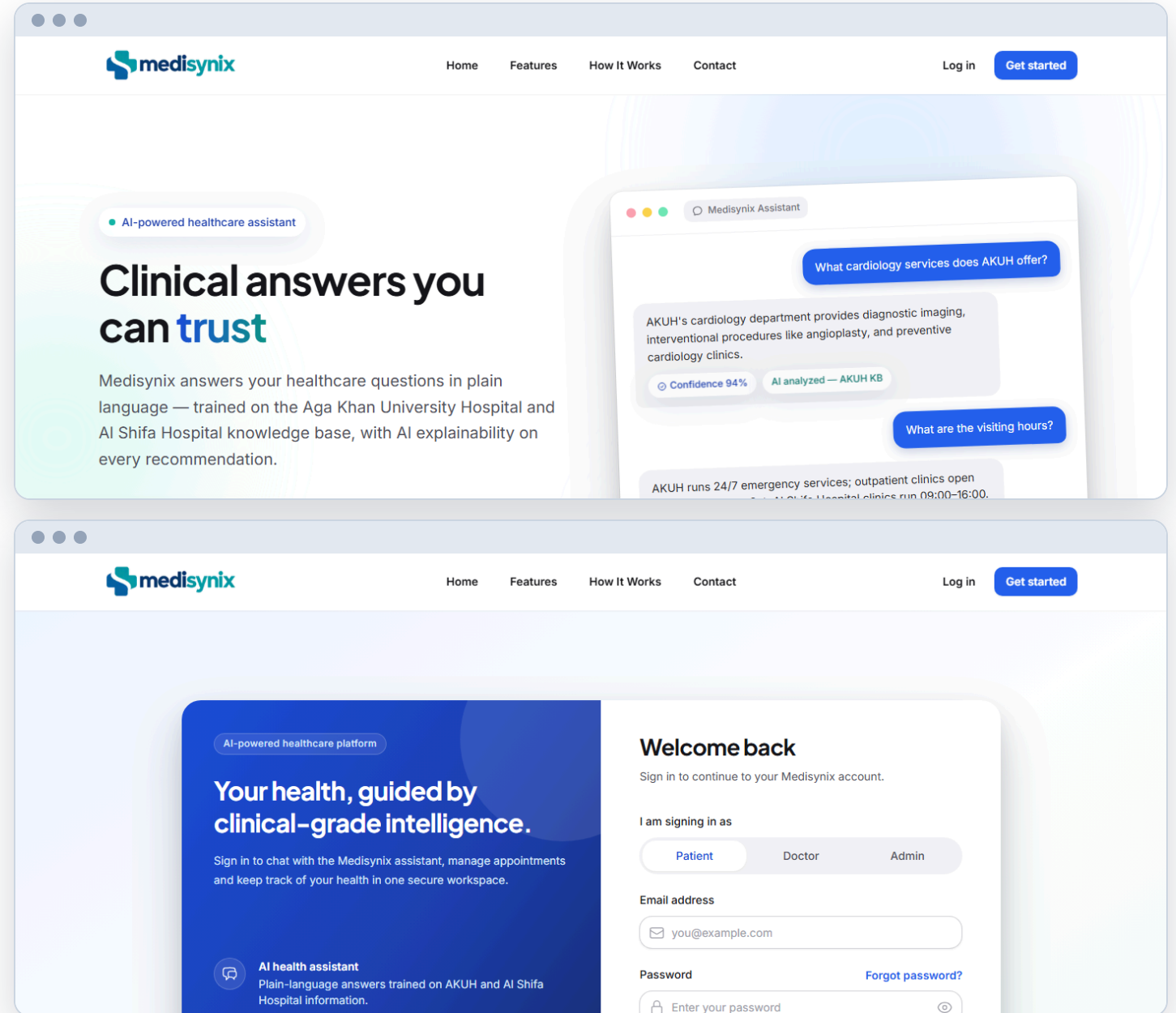
DO THIS

- Open **localhost:3000**. Scroll slowly: Features → How it works → Contact.
- Click **Log in**. Keep the **Patient** tab selected.
- Sign in as **ayesha.khan@demo.medisynix.test / Demo@1234**.

Say: This is Medisynix. Anyone can read about it, but health data is only visible after logging in, and each role gets its own workspace.

BEHIND THE SCENES

The login page checks the password (hashed with bcrypt) and the chosen role, then issues a signed token valid for 7 days. A wrong password shows "Invalid credentials". New users can register from *Create an account*.



The dashboard: your health at a glance

One screen with vitals, status colours and shortcuts.

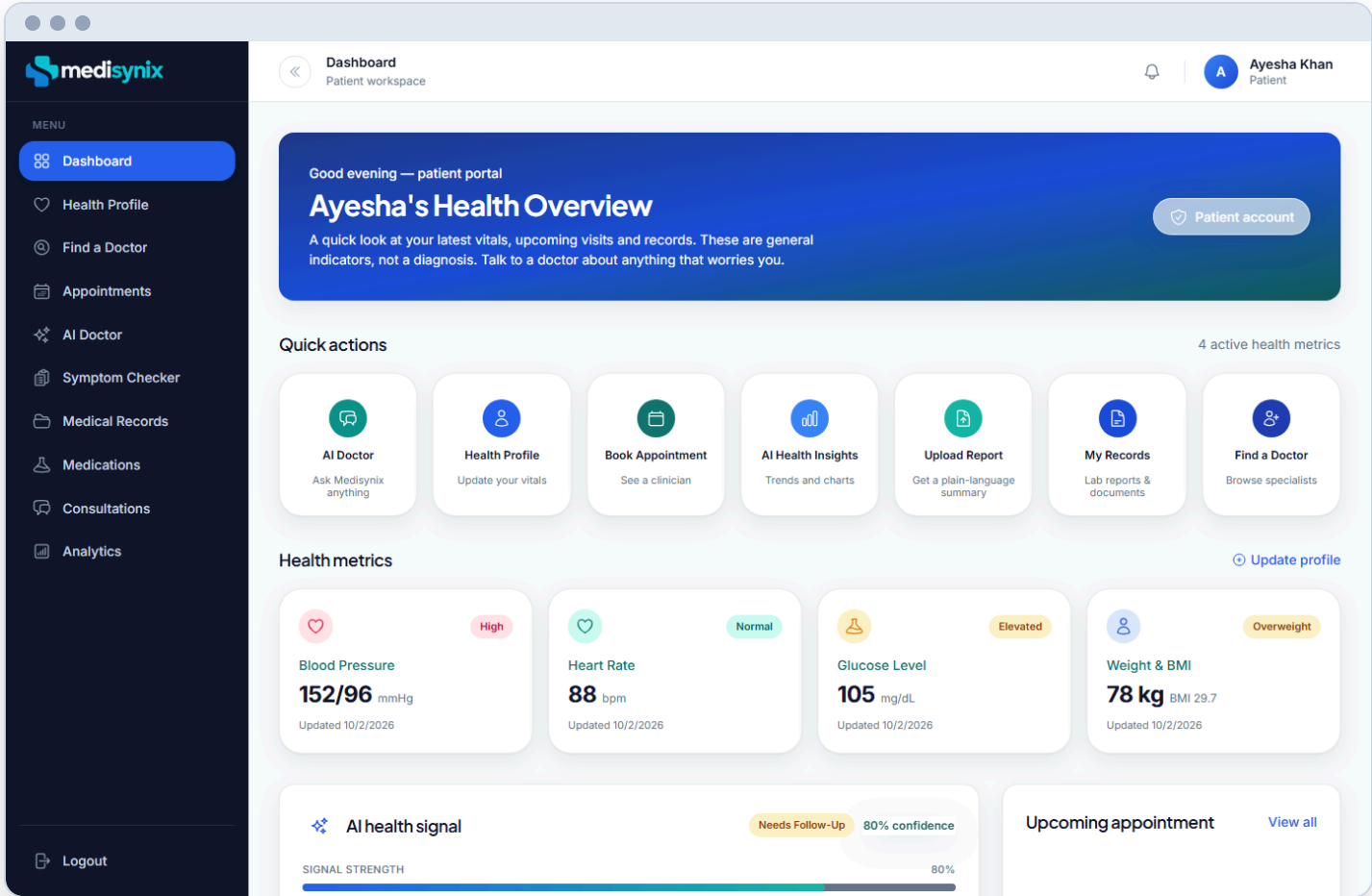
DO THIS

- Point at the **Quick actions** row, then the four **Health metrics** cards.
- Notice the colour tags: High (red), Normal (green), Elevated (amber).
- Scroll to the **AI health signal** and the **Upcoming appointment** card.

Say: Ayesha's blood pressure is 152/96 and the app flags it as High straight away, so she knows to act, without anyone reading a chart for her.

BEHIND THE SCENES

Cards are calculated from her latest saved readings with standard ranges (for example 140/90 or higher is flagged). The "AI health signal" is a rule-based score, not a diagnosis.



Health profile and vitals

Keep personal details and measurements up to date.

DO THIS

- Sidebar → **Health Profile** → **Edit Profile**.
- Show the fields: contact, date of birth, blood type, allergies, conditions, height, weight, BP, heart rate, glucose.
- Optional: type a wrong value (weight 9000) to show validation, then **Cancel**.

Say: Everything here is used later: by the dashboard, the AI, and by the doctor who treats her. Impossible values such as a 9,000 kg weight are rejected.

BEHIND THE SCENES

Saving calls the profile API which validates ranges (height 50–260 cm, weight 2–500 kg, heart rate 20–250, glucose 20–900, BP like 120/80), then stores one history entry per update. BMI is calculated automatically.

medisynix

MENU

Dashboard

Health Profile

Find a Doctor

Appointments

AI Doctor

Symptom Checker

Medical Records

Medications

Consultations

Analytics

Logout

Health Profile

Patient workspace

Ayesha Khan

Patient

Patient Profile

Cancel

Personal Information

Full Name

Ayesha Khan

Email

ayesha.khan@demo.medisynix.test

Phone Number

Date of Birth

mm/dd/yyyy

Gender

Select Gender

Blood Type

Select Blood Type

Health Metrics

Height (cm)

162

Weight (kg)

78

BMI: 29.7 - Overweight

Find a doctor

Search by name, speciality or place and see real time slots.

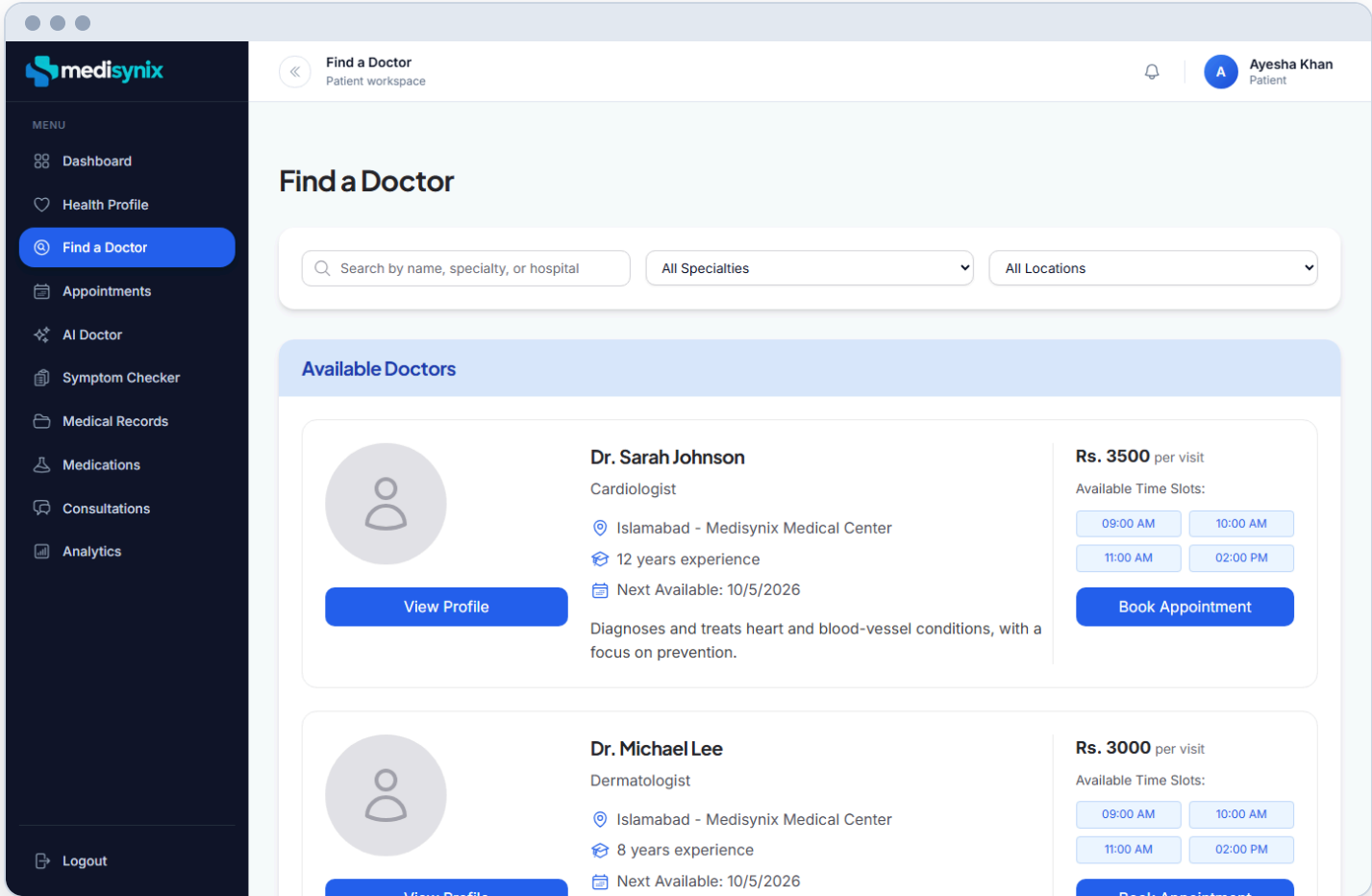
DO THIS

- Sidebar → **Find a Doctor**.
- Type **cardio** in the search box, or use the speciality / location filters.
- Click **View Profile** on Dr. Sarah Johnson.

Say: Each doctor shows fee, experience, location and open slots. The slots come from the doctor's own profile, so patients can only choose times the doctor offers.

BEHIND THE SCENES

The directory API merges doctors created by the admin. A booked slot disappears for others on that date, and cancelling a booking frees it again.



Book an appointment

From profile to confirmed booking in three clicks.

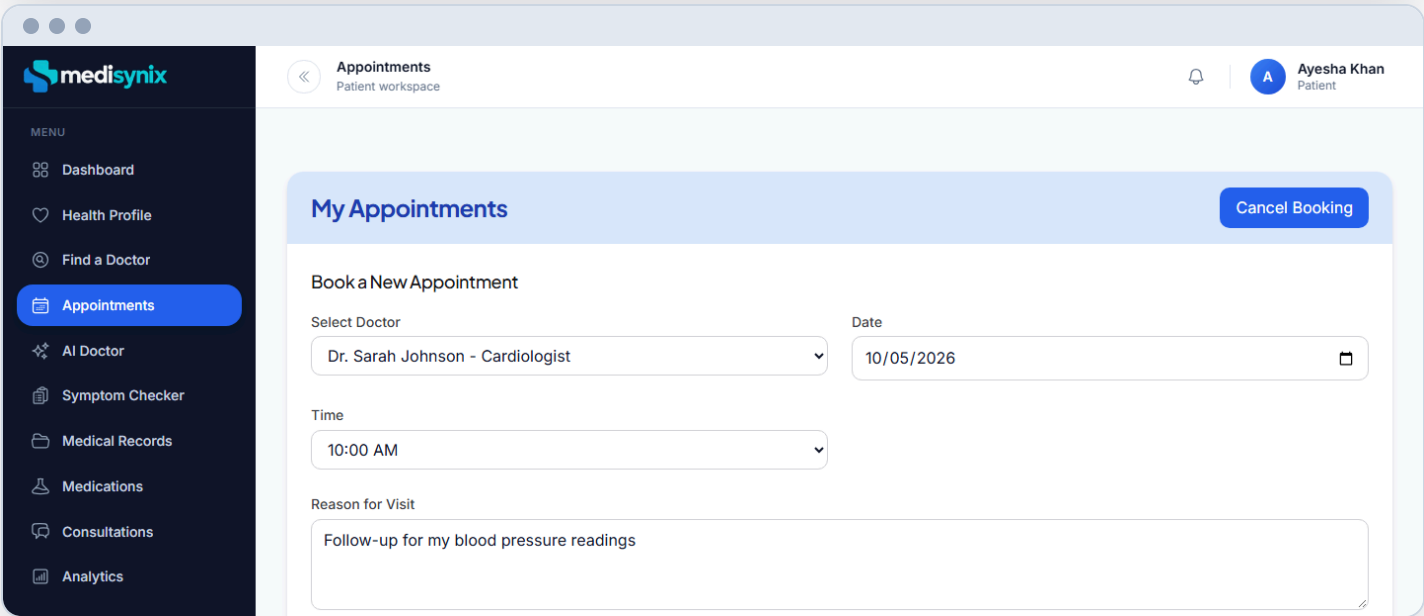
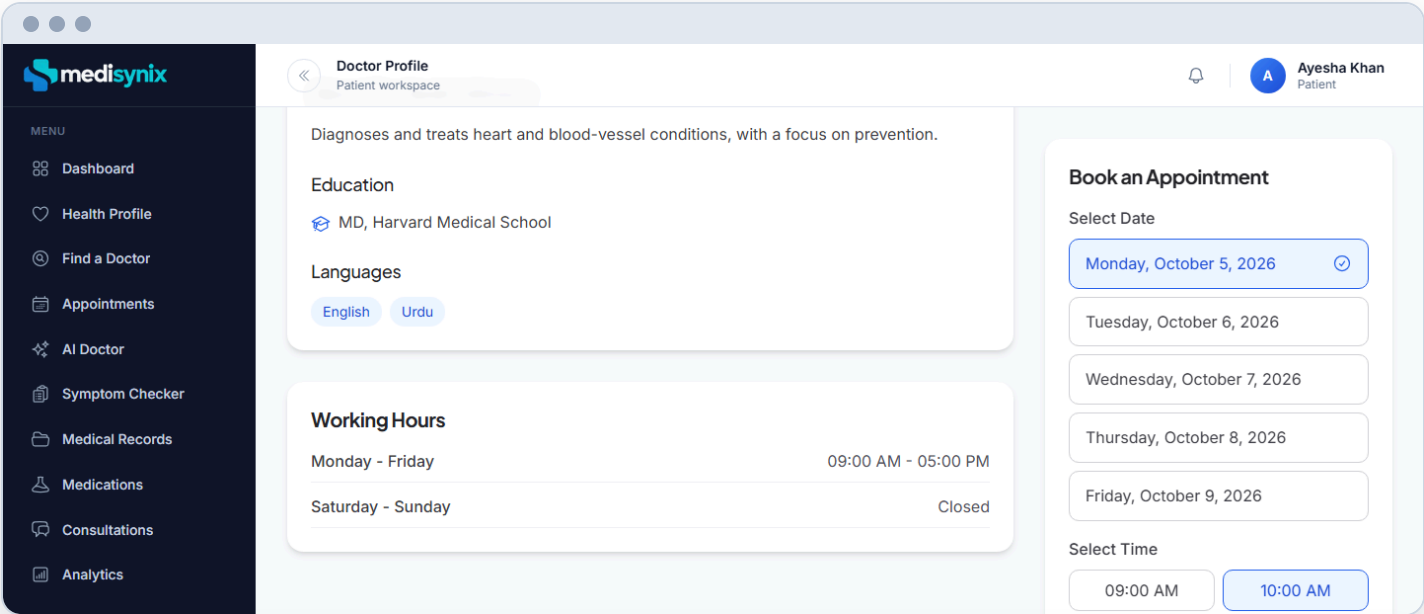
DO THIS

- On the doctor page pick a **date** and a **time slot** (the button becomes active).
- Click **Book Appointment**, add a reason, confirm.
- Sidebar → **Appointments**: the visit appears under Upcoming. Show the Past and Cancelled tabs.

Say: If two patients try the same slot, the second one gets a clear message instead of a double booking. Past dates are refused too.

BEHIND THE SCENES

Server rules: no double bookings (HTTP 409), no past or invalid dates (400), only offered time slots. Cancelling returns the slot to the pool. The doctor is notified on their bell.



AI Doctor (real language model)

Ask questions in everyday language.

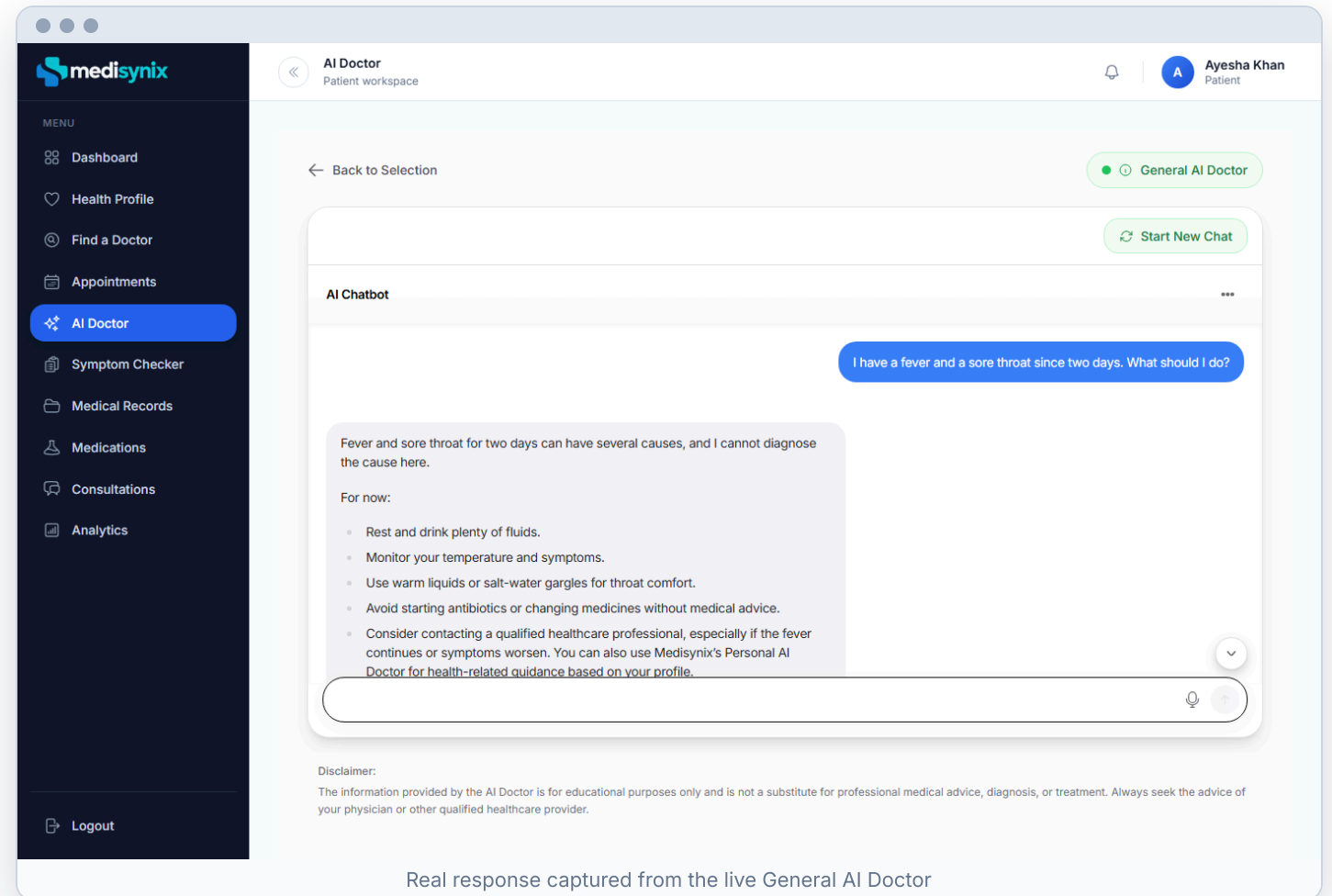
DO THIS

- Sidebar → **AI Doctor** → **Select General AI Doctor**.
- Type: *"I have a fever and a sore throat since two days. What should I do?"*
- Optional: go back and open the **Personal AI Doctor**, which shows her saved health summary beside the chat.

Say: The answer is general guidance, never a diagnosis: rest, fluids, watch the temperature, and when to see a doctor. It also points to the Personal AI Doctor.

BEHIND THE SCENES

Powered by a Chatbase agent trained on Aga Khan University Hospital and Al Shifa Hospital information. The Personal view shows the profile on screen; it does not send her record to the model.



Real response captured from the live General AI Doctor

Symptom checker with emergency warnings

A fast, transparent way to see which common conditions fit.

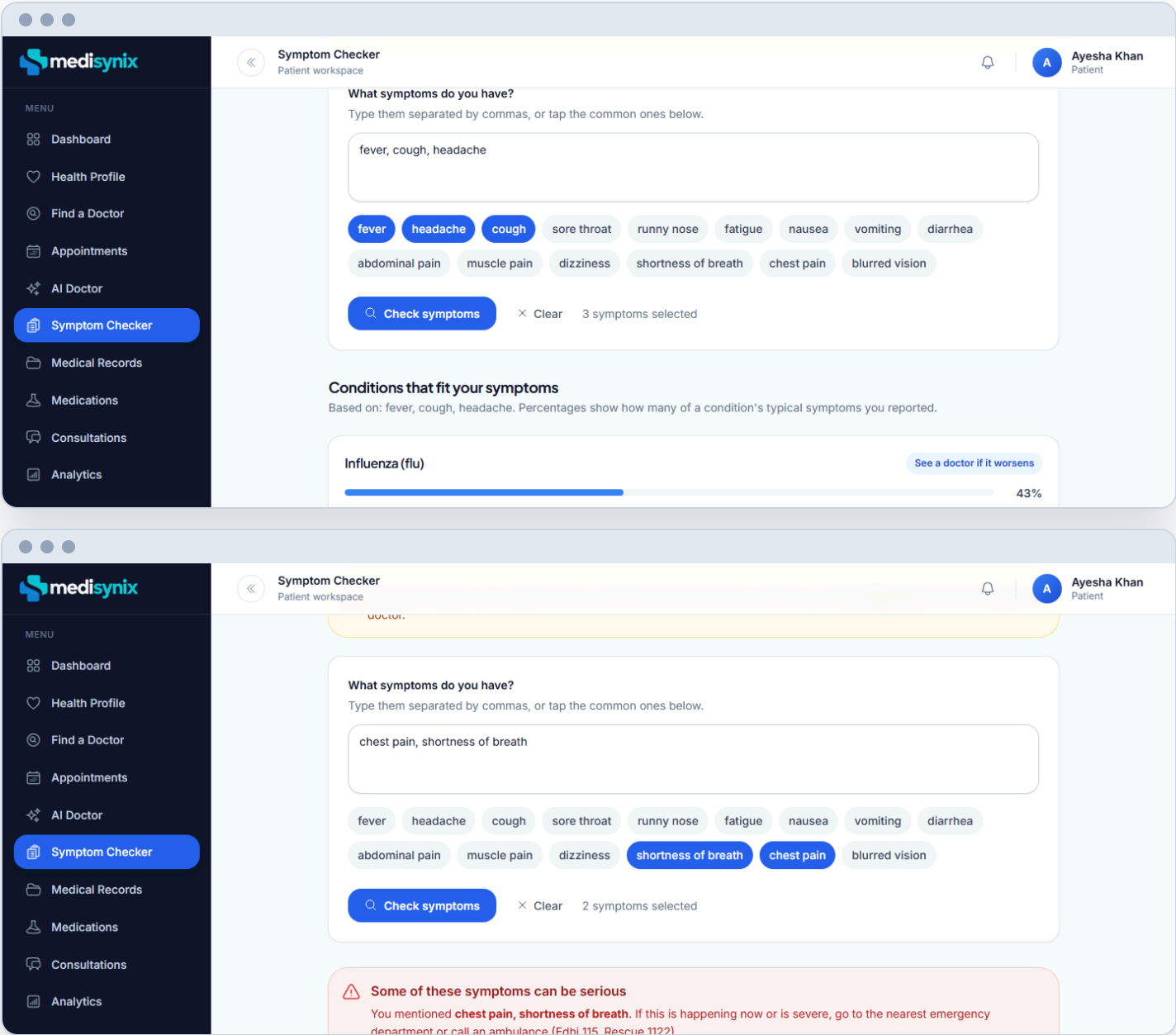
DO THIS

- Sidebar → **Symptom Checker**.
- Tap chips **fever**, **cough**, **headache** → **Check symptoms**. Show the % match bars and advice.
- **Clear**, then tap **chest pain** and **shortness of breath**: a red emergency alert appears (Edhi 115, Rescue 1122).

Say: It tells you how many of a condition's typical symptoms you reported, what to do, and when to seek urgent help. It is a guide, not a diagnosis, and says so.

BEHIND THE SCENES

A small built-in table of 10 common conditions. Matching is by symptom phrase; red-flag phrases such as chest pain or coughing blood trigger the emergency banner independently of the matches.



Upload a report and get a plain-language summary

Turn confusing lab or imaging reports into something readable.

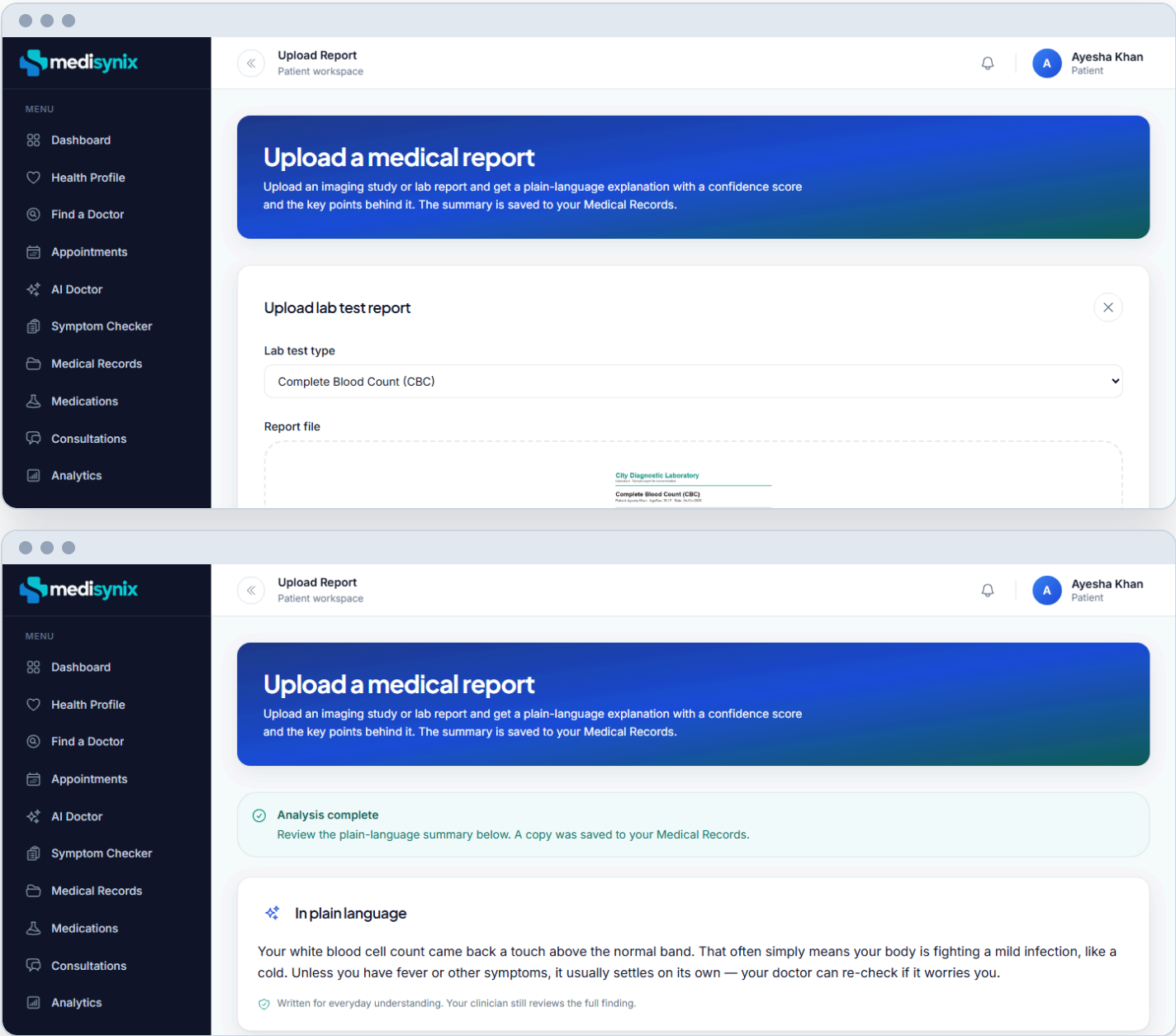
DO THIS

- Dashboard → **Upload Report** (or sidebar **Medical Records** → **Add record**).
- Choose **Lab test report**, select a test (for example CBC), attach any image or PDF, click **Upload and analyze**.
- Show **In plain language**, the table with High / Normal tags, and **How Medisynix reached this reading** (confidence + reasoning steps).

Say: The idea is explainable AI: not just an answer, but how sure the system is and why. The summary is saved to her records automatically.

BEHIND THE SCENES

This is a prototype: the explanation is a sample for the selected test type, and the page says so. The roadmap connects it to a medical model that reads the file.



Medical records and medicines

A tidy personal health file.

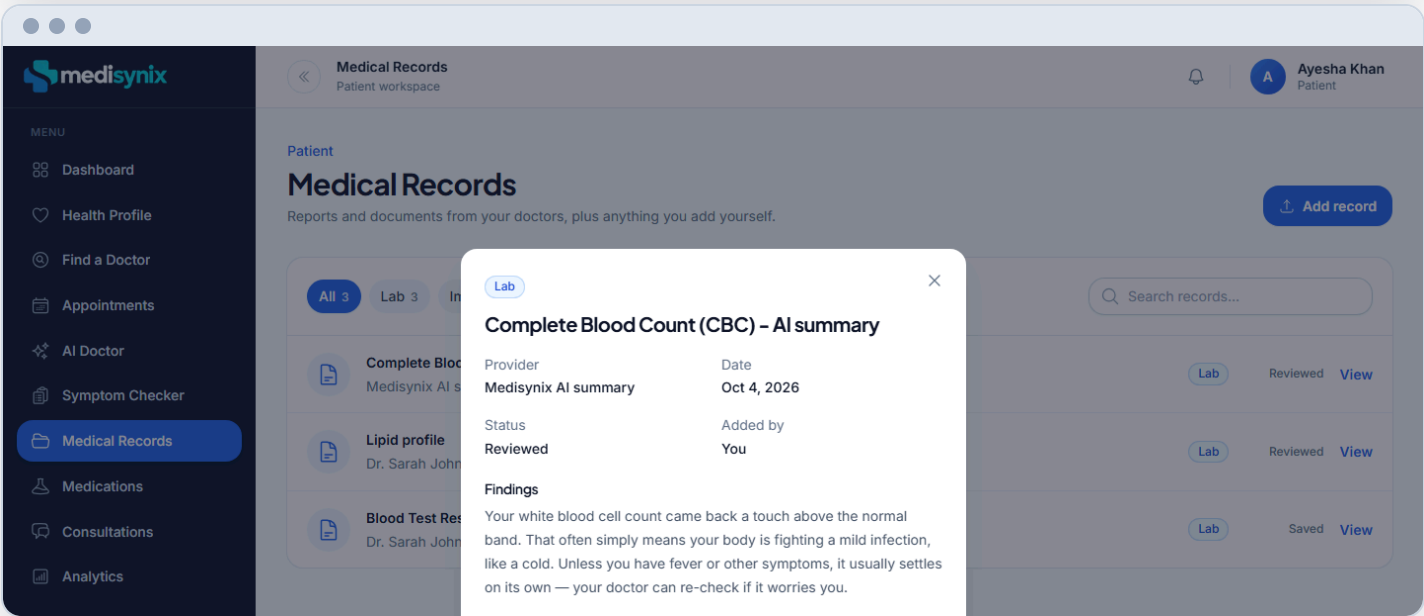
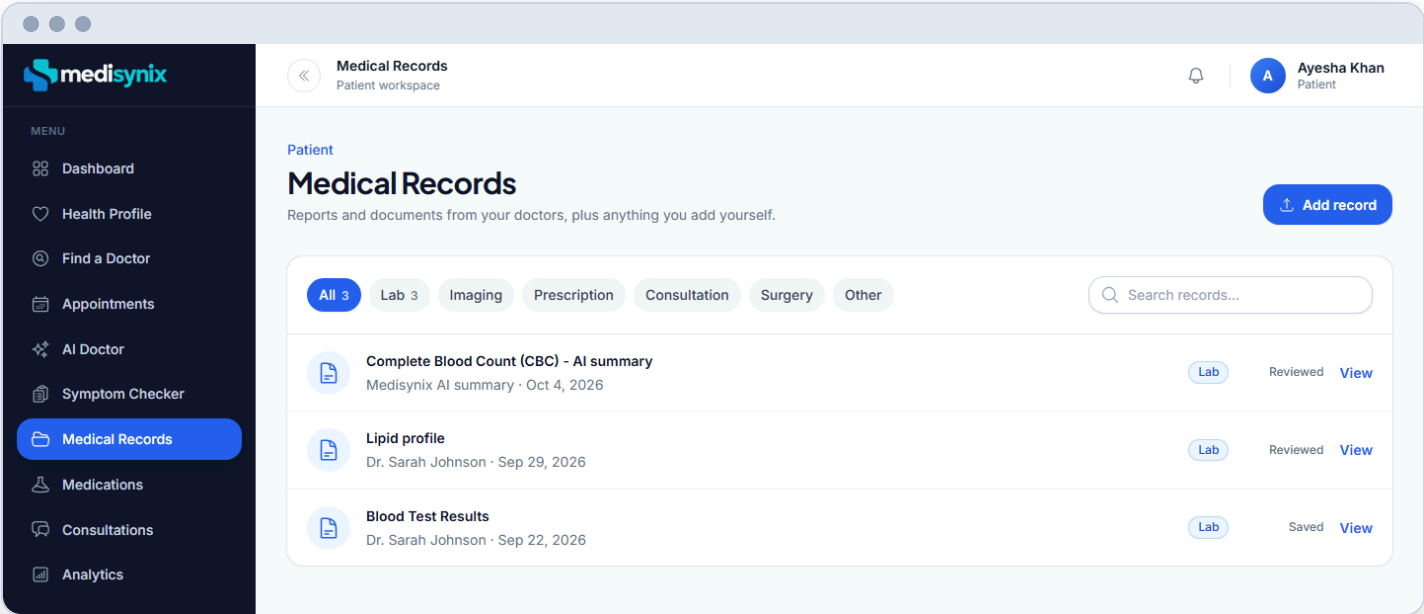
DO THIS

- Sidebar → **Medical Records**: use the type chips and search, click **View** to open a record, show Delete on her own record.
- Sidebar → **Medications**: show today's reminders and the table; **Add Medication** then **Remove**.

Say: Records from doctors and from the patient live together. Patients can add or delete their own, but cannot change what a doctor uploaded.

BEHIND THE SCENES

Doctor-uploaded reports arrive here automatically with the doctor's name. Attachments up to 2 MB are stored with the record. Medicines are saved per patient.



Visits and health insights

See care history and trends.

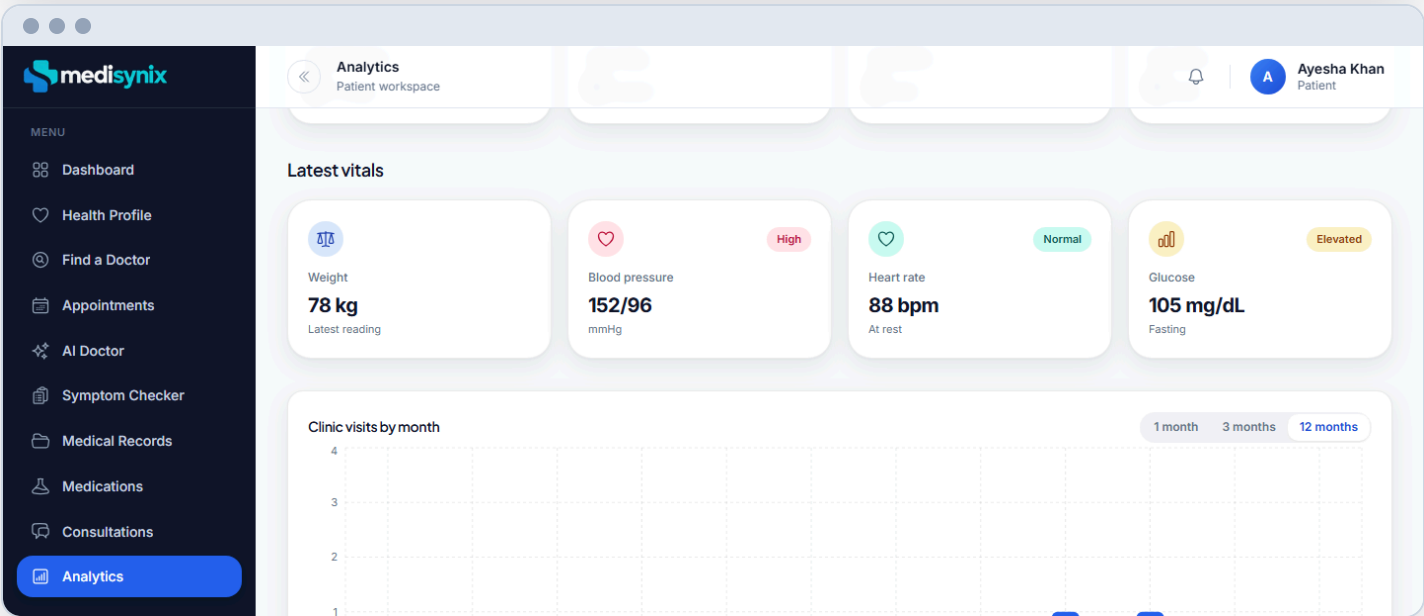
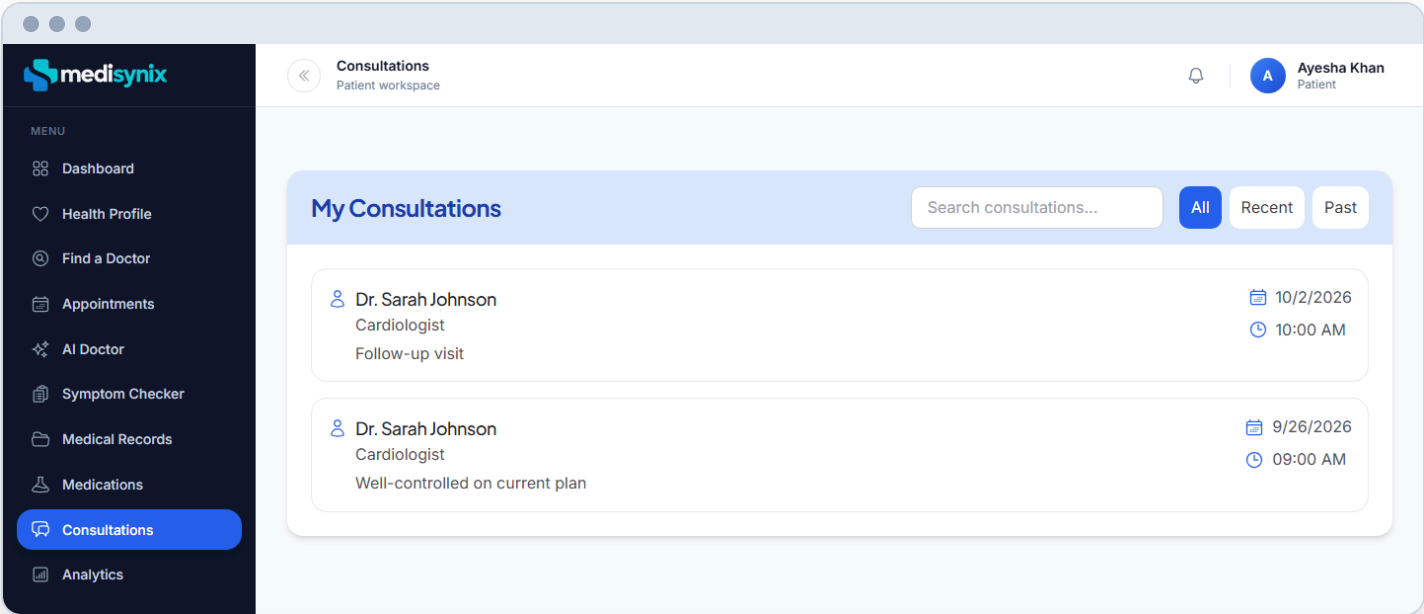
DO THIS

- Sidebar → **Consultations**: open the completed visit to read the doctor's diagnosis and notes.
- Sidebar → **Analytics**: change the range (1 / 3 / 12 months) and show the charts.
- Finish by clicking the avatar → **Logout**.

Say: What the doctor wrote after the visit appears here for the patient: the loop is closed. The insights page only uses real data from her account.

BEHIND THE SCENES

Consultations are the completed appointments. Analytics counts her appointments and plots her saved vitals over time; if she has none, a helpful empty state appears instead of fake numbers.



DOCTOR

Dr. Sarah Johnson, Cardiologist

Sees patients through Medisynix next to a clinic. Opens the app before the first visit and again between consultations.

What this person wants: Know who is coming today, spot patients who need attention, record each visit quickly and keep her public profile correct.

1 Sign in

2 Dashboard

3 Patients

4 AI analysis

5 Consultations

6 Reports & profile

Sign in and start the day

A live overview of the doctor's workload.

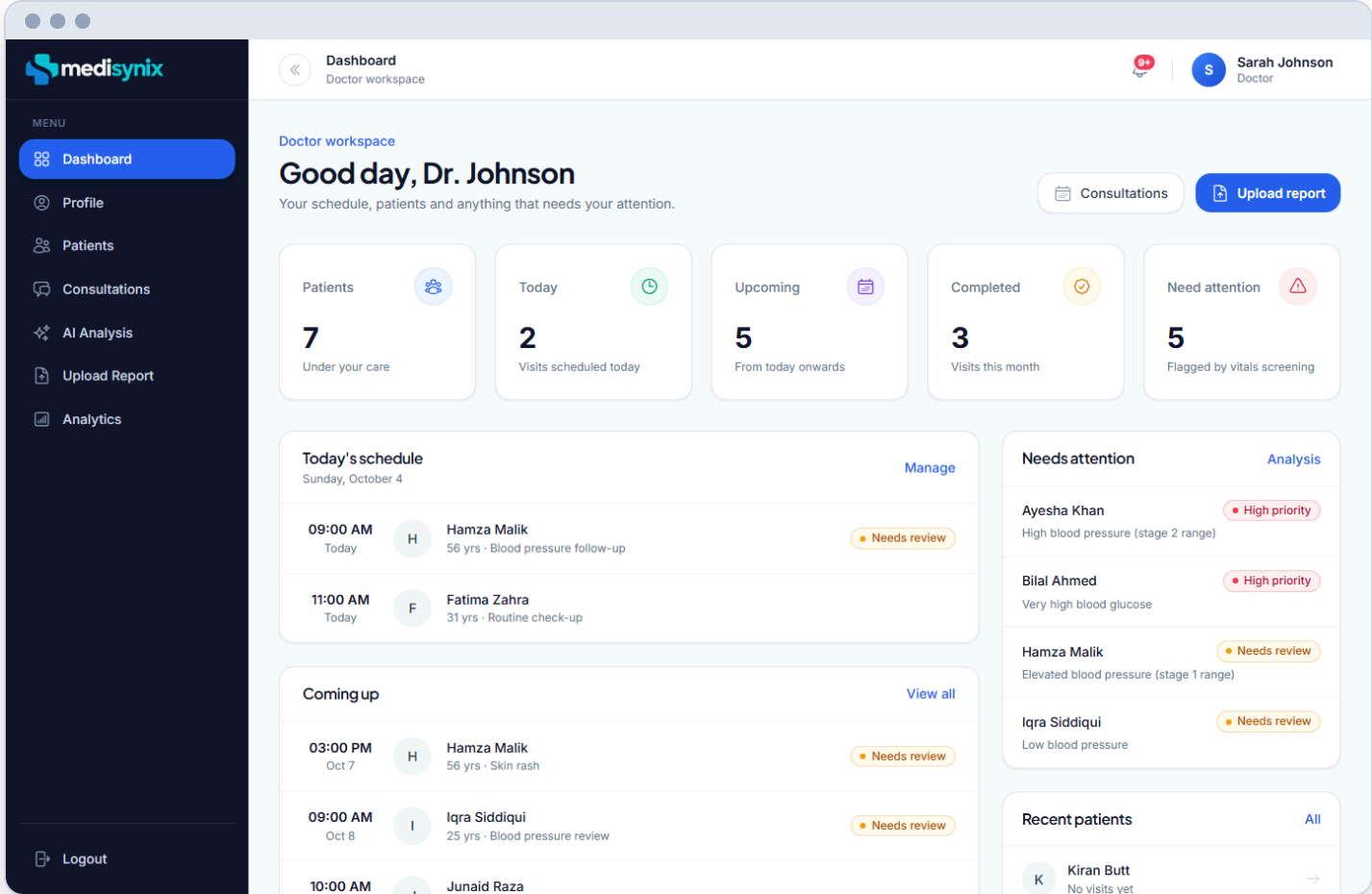
DO THIS

- In the Doctor browser tab: **Log in** → tab **Doctor** → **dr.sarah@medisynix.com / doctor123**.
- Read the five counters: Patients, Today, Upcoming, Completed, Need attention.
- Show **Today's schedule**, **Coming up** and **Needs attention**. Click the bell to see new bookings.

Say: A patient booked yesterday and the doctor sees it today. Names, times and reasons are real; the list of people who need attention is generated from vitals.

BEHIND THE SCENES

The overview API only returns patients who have booked with this doctor. If past visits are still open, an amber banner reminds the doctor to complete or cancel them.



Patients and the patient file

Everything about one patient on a single page.

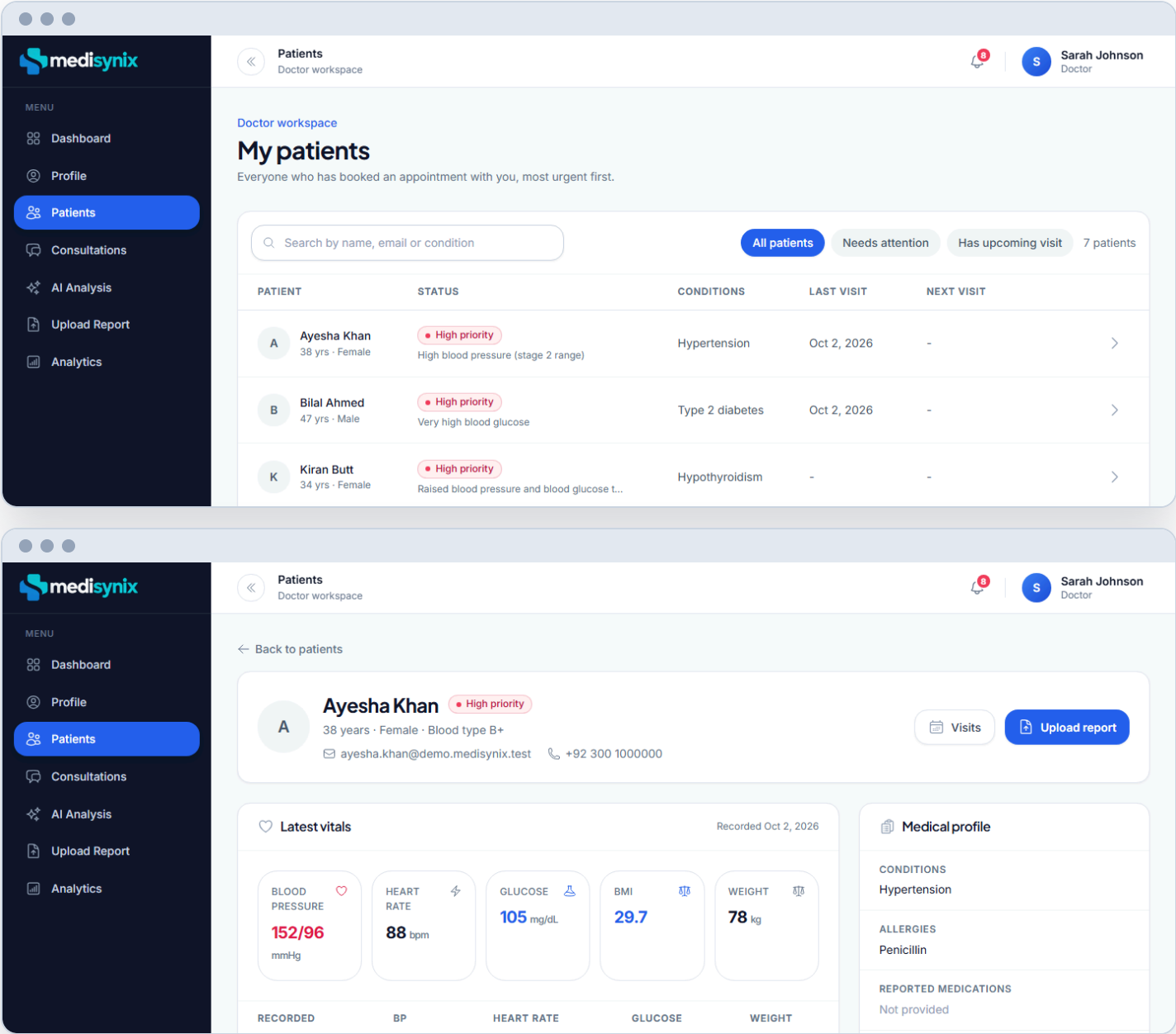
DO THIS

- Sidebar → **Patients**: search, use **Needs attention** filter.
- Click **Ayesha Khan**: latest vitals, history, screening flags, medical profile, medicines, records.
- Show the **Upload report** button on this page.

Say: No more hunting through papers: vitals are colour-coded, and a reading like 152/96 is explained as 'stage 2 range' with a suggestion to confirm with repeat readings.

BEHIND THE SCENES

A doctor can open only patients who booked with them; any other patient looks like 'not found'. This is enforced on the server, not just hidden in the interface.



AI analysis: who needs attention first?

Rank patients by how urgently their vitals deserve a look.

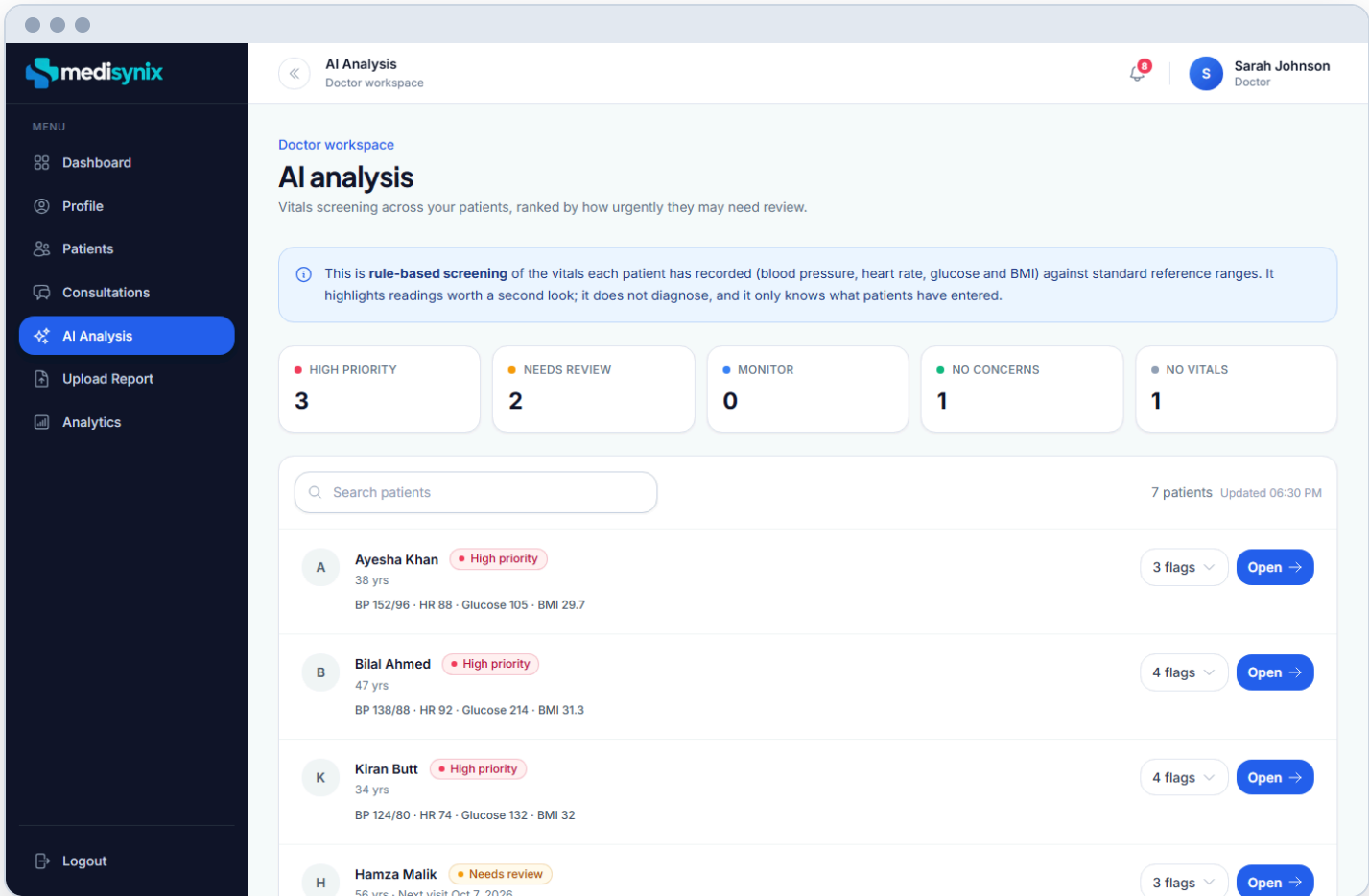
DO THIS

- Sidebar → **AI Analysis**.
- Click the **High priority** tile to filter. Expand a patient with **flags** to see the reading and reason.
- Click **Open** to jump to that patient's file.

Say: This is decision support, not a verdict. The banner on the page says it is rule-based screening of what patients entered. It tells the doctor where to look first.

BEHIND THE SCENES

Rules compare blood pressure, heart rate, glucose and BMI to reference ranges. Combined flags raise priority. The same logic powers the 'Needs attention' panel on the dashboard.



Consultations: complete, reschedule or cancel

Run the day and record what happened.

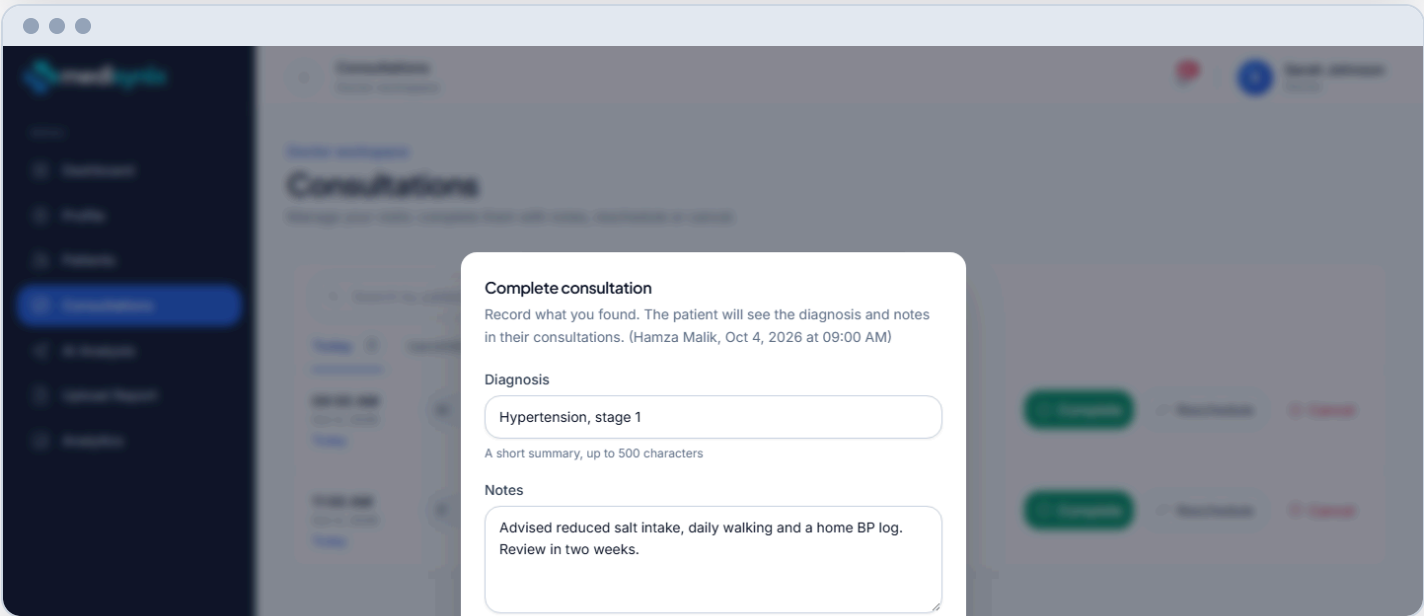
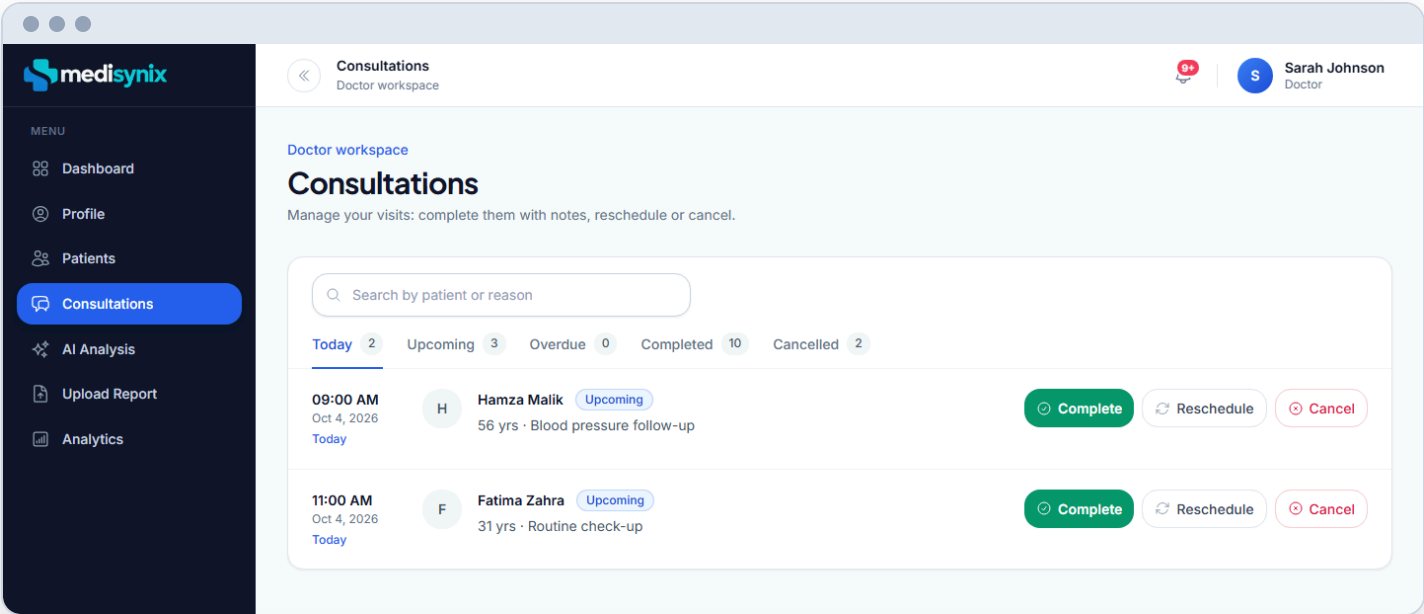
DO THIS

- Sidebar → **Consultations**. Tabs: Today, Upcoming, Overdue, Completed, Cancelled.
- In **Today** click **Complete** on a visit, type a diagnosis and notes, click **Mark completed**.
- Show **Reschedule** (only free slots) and **Cancel** (optional reason).

Say: The diagnosis and notes written here are exactly what the patient reads in their own Consultations page. One entry, both sides updated.

BEHIND THE SCENES

Rules: only today's or past visits can be completed, finished visits cannot be reopened, rescheduling to a taken slot is refused, and cancelled visits are final.



Send a report and review the practice

Deliver results to the patient and see the numbers.

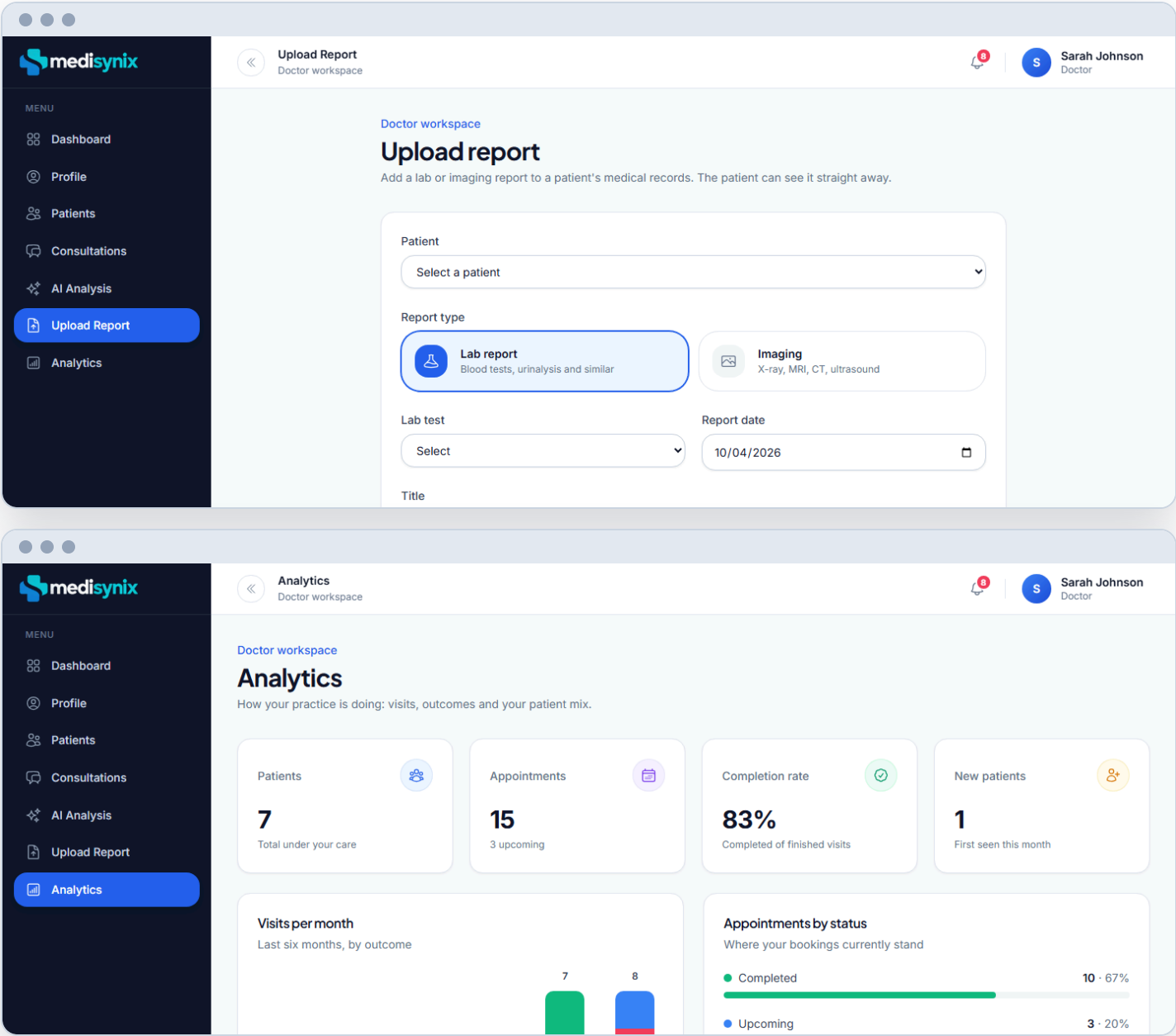
DO THIS

- Sidebar → **Upload Report**: choose the patient, lab or imaging, test, title, date, findings, optional PDF/PNG/JPEG (up to 2 MB).
- Sidebar → **Analytics**: visits per month, outcomes, busiest weekdays, top reasons.

Say: The doctor's findings are typed by the doctor. Nothing is auto-generated. The patient sees the report in their records with the doctor's name.

BEHIND THE SCENES

Validation refuses unknown tests, missing findings, future dates, wrong file types and files over 2 MB. Analytics are calculated from real appointments only.



Public profile and availability

Control what patients see and when they can book.

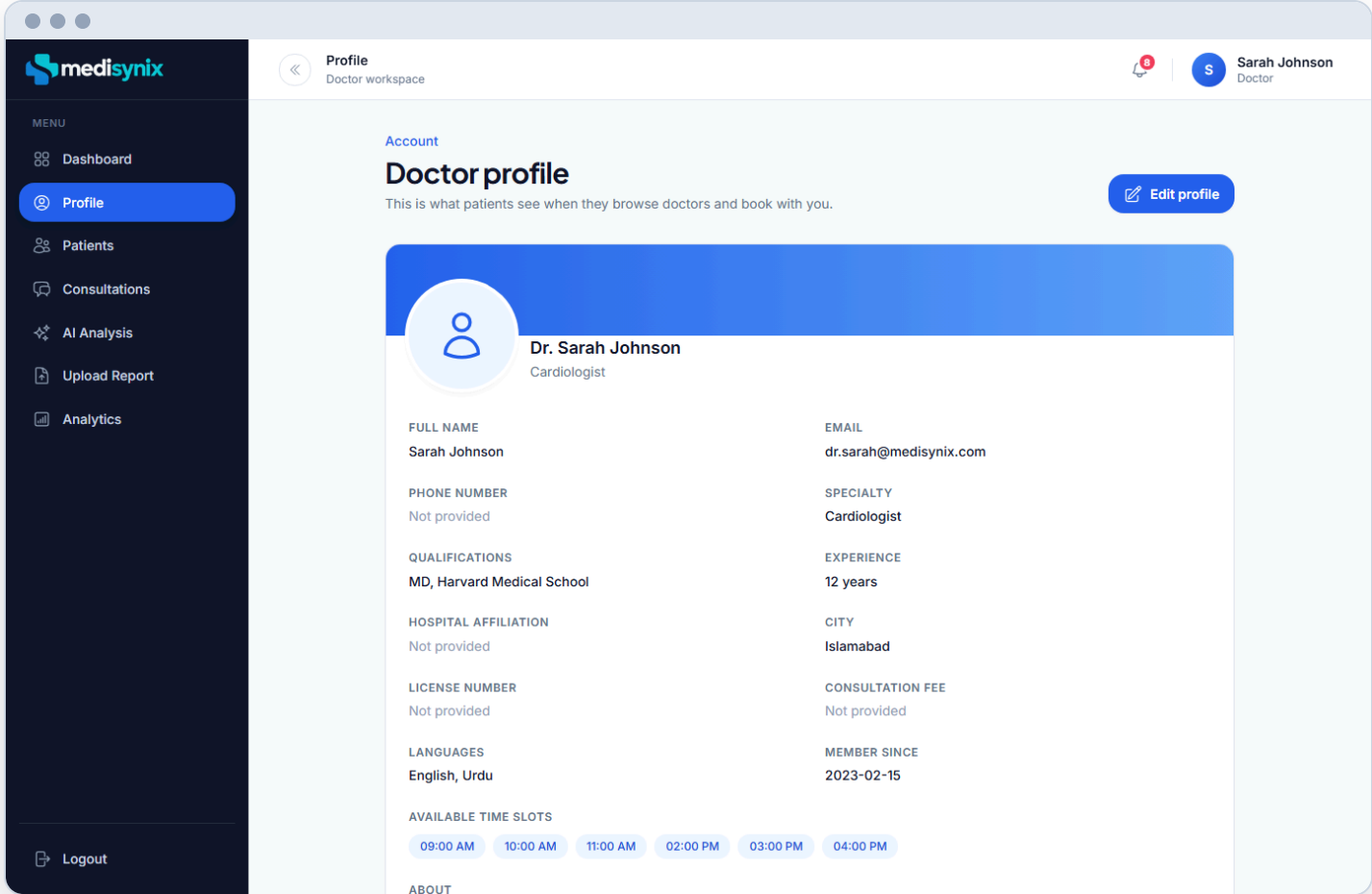
DO THIS

- Sidebar → **Profile**.
- Show specialty, city, qualifications, fee, languages and the **available time slots** chips.
- Optional: remove a slot, save, then show that the patient booking form no longer offers it.

Say: A change here is instantly reflected in Find a Doctor and in the booking form. The doctor owns their schedule.

BEHIND THE SCENES

Dropdowns restrict values (specialty, city). Invalid phone numbers, fees or empty slot lists are rejected. The photo is stored with the profile.



ADMINISTRATOR

Mohsin Furkh, Platform Admin

Owns the platform's accounts and health. Signs in a few times a day to look for problems and onboard doctors.

What this person wants: See what needs attention in seconds, manage accounts safely, and be stopped before doing something destructive.

1 Sign in & dashboard

2 Manage users

3 Health & security

4 Reports & settings

Admin dashboard and system status

Platform health in one glance.

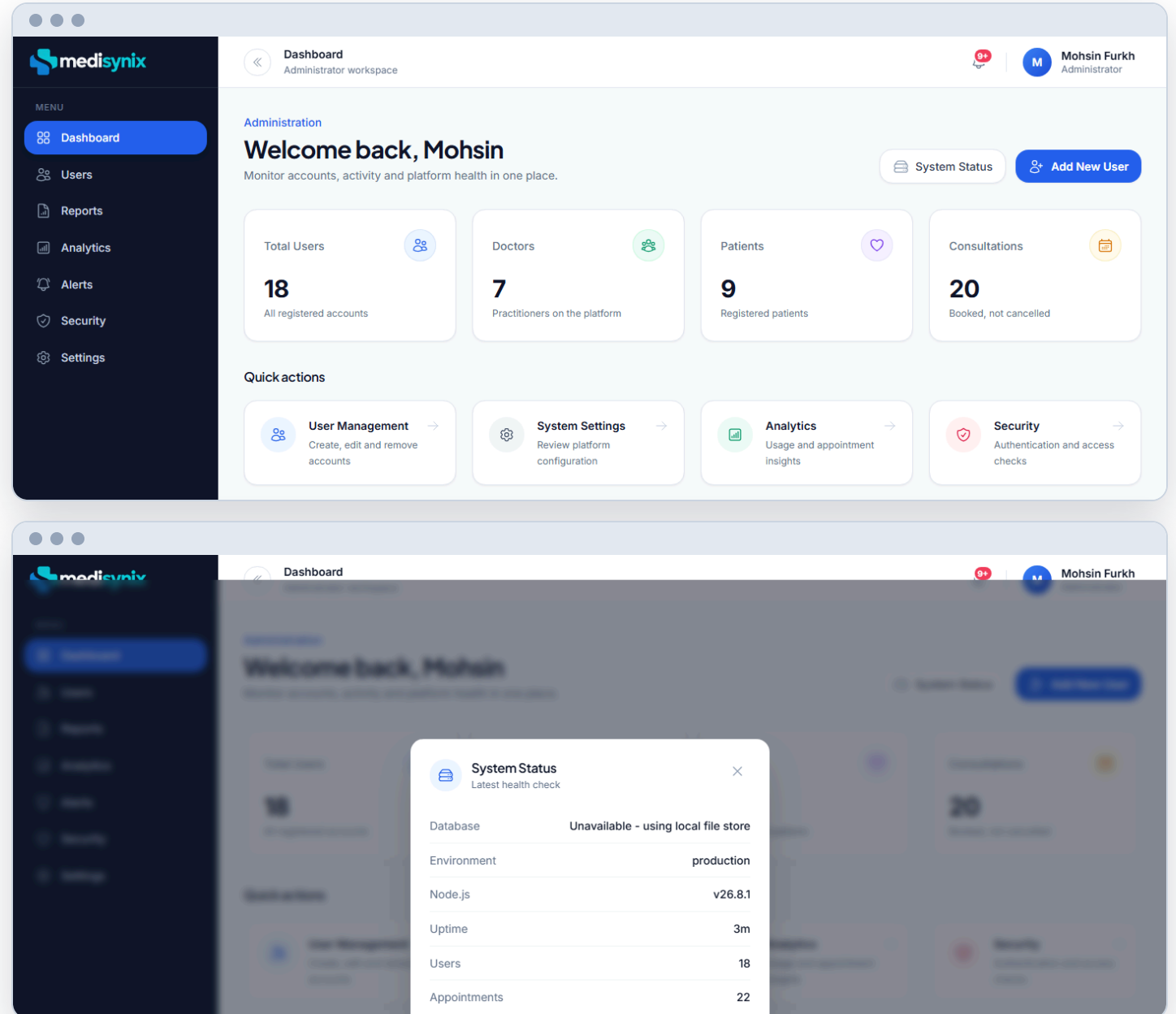
DO THIS

- In the Admin tab: **Log in** → tab **Admin** → **admin@medisynix.com / admin123**.
- Read the counters: users, doctors, patients, consultations. Click the tabs **Recent Users / System Alerts / Reports**.
- Click **System Status** to open the status dialog.

Say: The administrator manages accounts and platform health, not clinical records. Numbers here always match the Users and Analytics pages.

BEHIND THE SCENES

System Status shows database mode (database or local files), environment, uptime and totals. Role guards: a doctor or patient calling any admin API gets 403, a forged token gets 401.



User management

Create, find, edit and remove accounts safely.

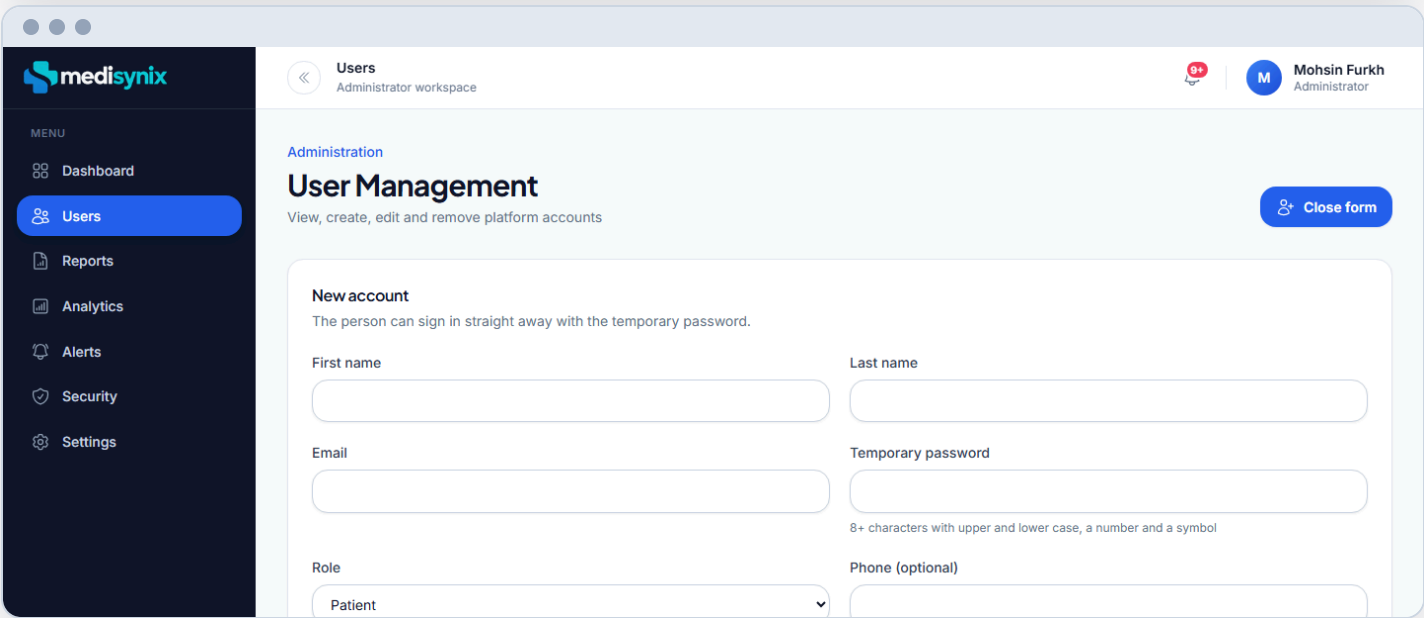
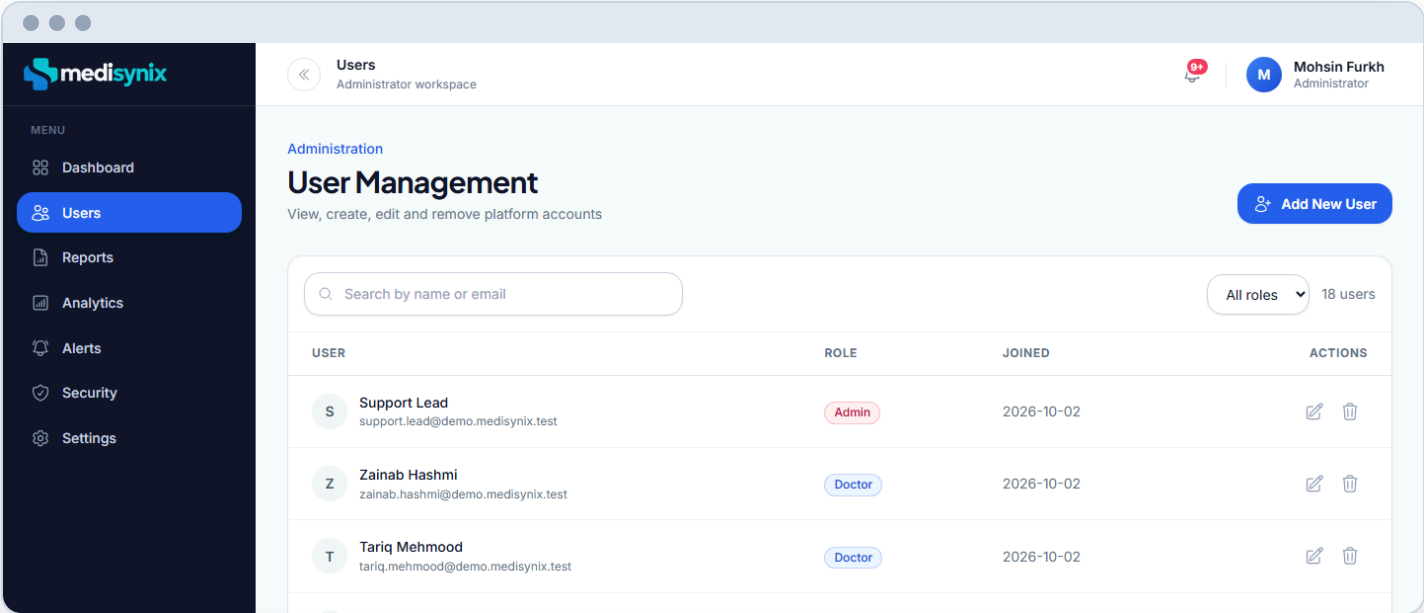
DO THIS

- Sidebar → **Users**. Search by name or e-mail, filter by role.
- Click **Add New User**: fill the form (patient, doctor or admin) and **Create account**. Show the weak-password message first.
- Open the edit icon, then show the delete confirmation (do not confirm).

Say: A new doctor appears immediately in Find a Doctor. Deleting someone also removes their appointments, records and medicines so statistics stay correct.

BEHIND THE SCENES

Guard-rails: duplicate e-mails refused, passwords need upper/lower case, a number and a symbol, admins cannot delete or demote themselves, and the last administrator cannot be removed.



Security, alerts and analytics

Know what is healthy and what needs fixing.

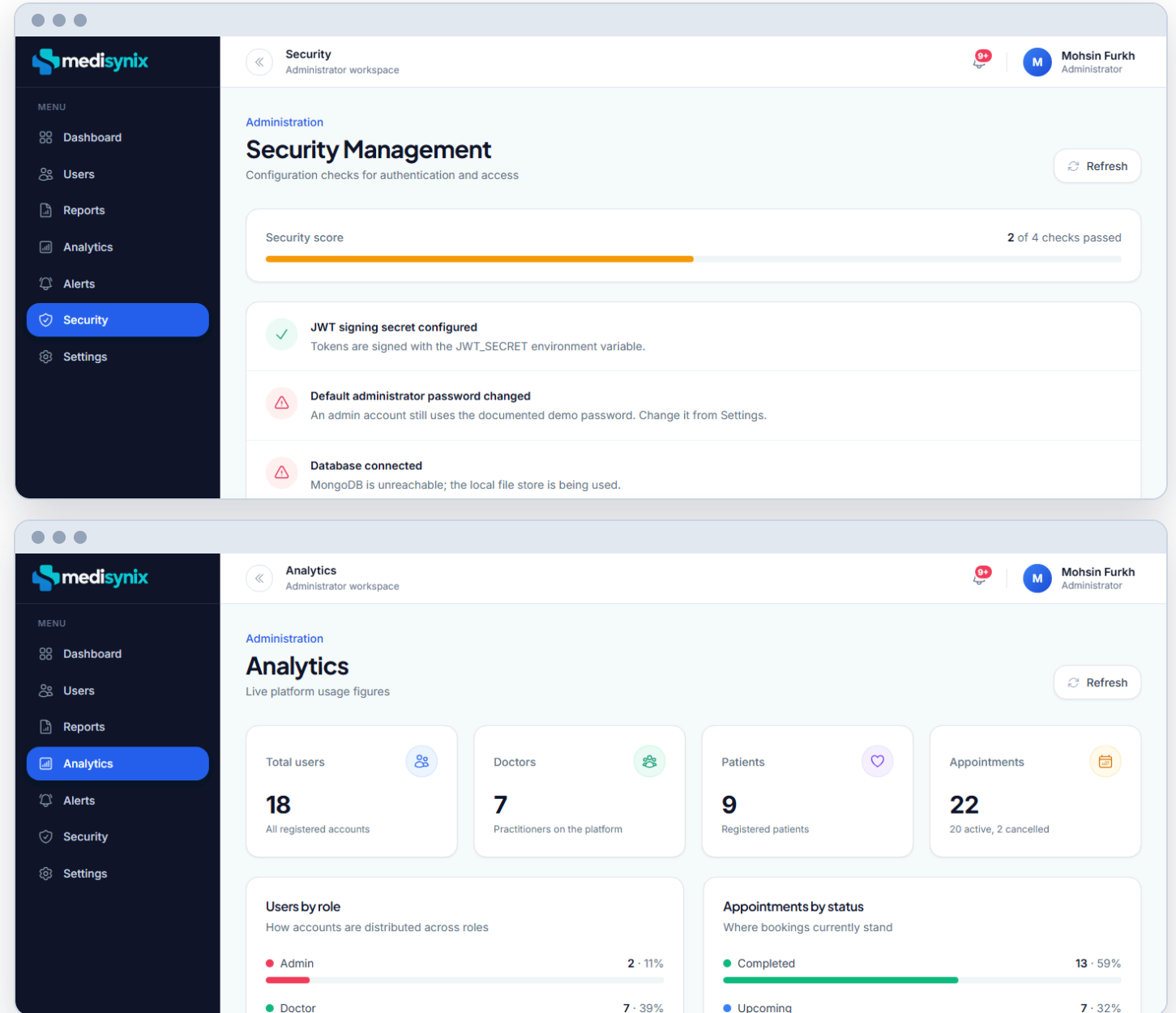
DO THIS

- Sidebar → **Security**: score of 4 checks. Explain the red items.
- Sidebar → **Alerts** and **Analytics**: users by role, appointments by status.
- Press **Refresh** to show it is live.

Say: Two checks fail honestly on this laptop: the default admin password is still in use and MongoDB is not connected, so we run on local files. In production both would be fixed.

BEHIND THE SCENES

The page inspects the real configuration (JWT secret, default password, database connectivity, number of admins). It is not a mock-up.



Reports and settings

Export data and manage the admin's own account.

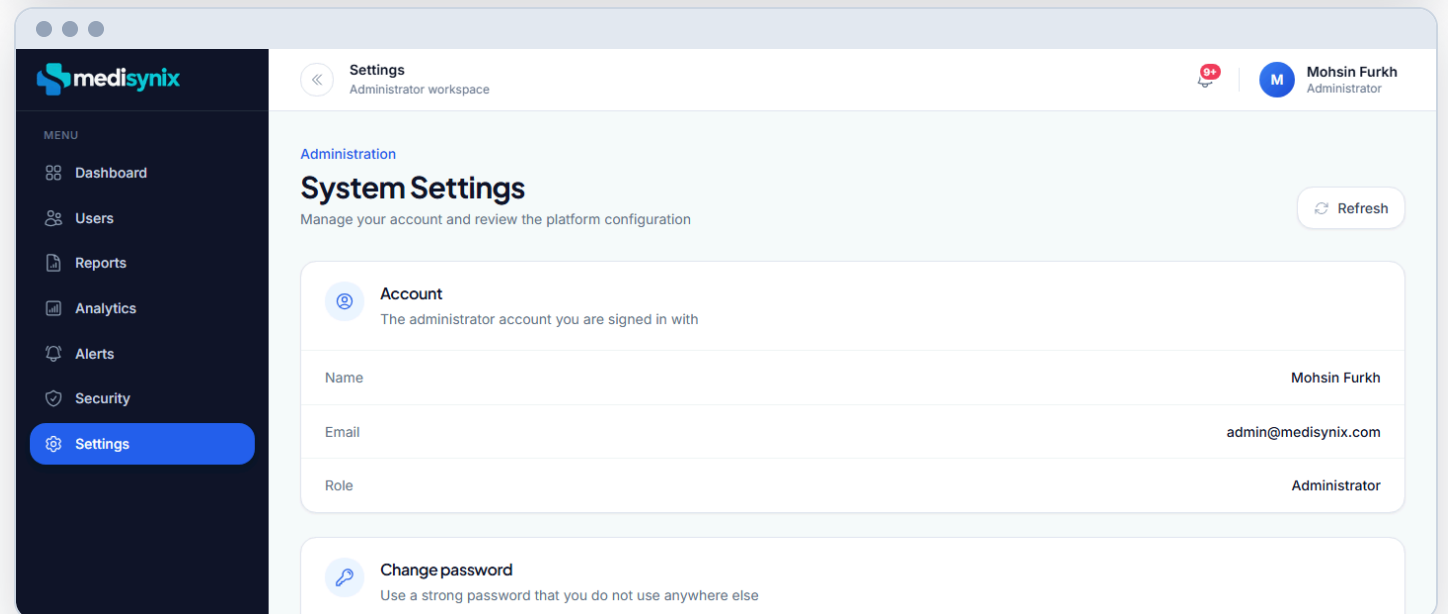
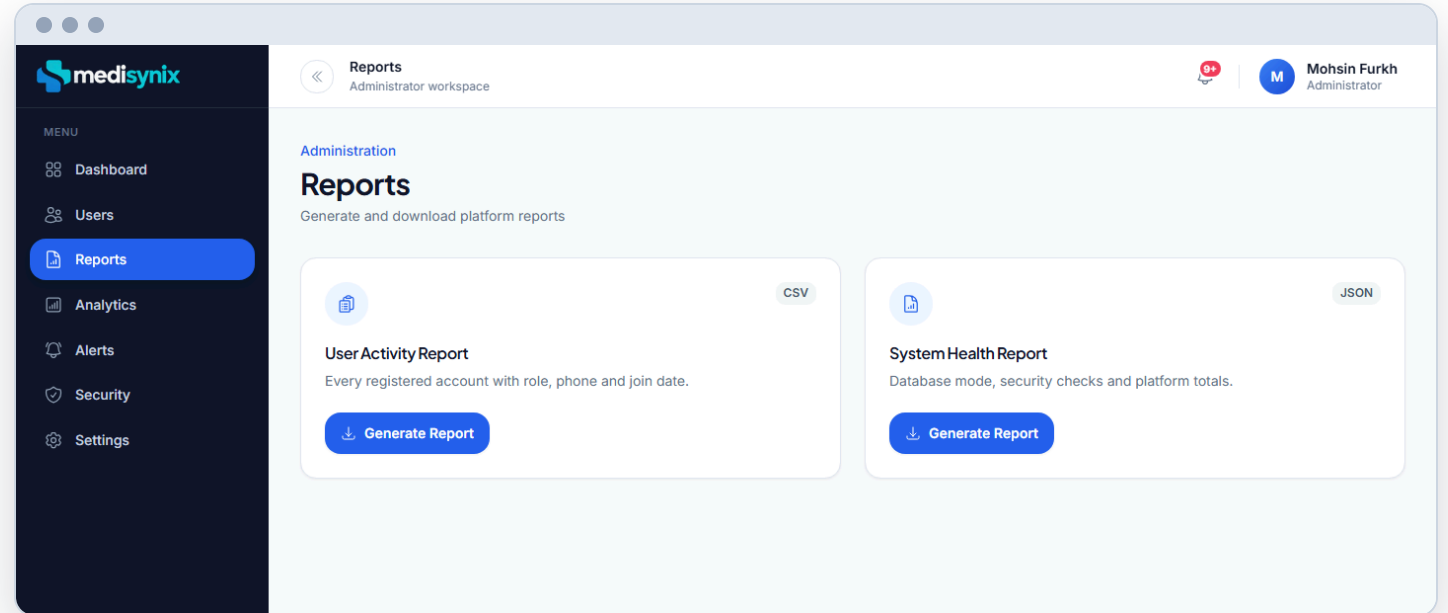
DO THIS

- Sidebar → **Reports**: click **User Activity Report** (CSV) and **System Health Report** (JSON).
- Sidebar → **Settings**: change-password form with live rule checklist.
- Click **Logout**: you are returned to the login page.

Say: Reports open in Excel or any tool. Spreadsheet formula characters in names are neutralised so a malicious name cannot run code.

BEHIND THE SCENES

Change password needs the current password and enforces the same rules as registration. Direct URLs such as /dashboard/admin redirect to login when signed out.



Closing the loop: what one role does, the other sees

Someone does this...	...and this person sees it
Patient books a slot	Doctor: shows in Coming up and the bell; slot disappears for other patients
Patient updates blood pressure to 152/96	Doctor: patient gets a High priority flag in AI Analysis and the dashboard
Doctor completes a visit with diagnosis and notes	Patient: Consultations page shows the diagnosis and notes; Appointments moves it to Past
Doctor cancels or reschedules	Patient: visit appears in Cancelled or with the new time
Doctor uploads a report	Patient: Medical Records shows it with the doctor's name
Doctor changes fee or time slots	Patient: Find a Doctor, doctor page and booking form update at once
Admin creates a doctor	Patients: the new doctor appears in Find a Doctor
Admin deletes a user	All their appointments, records and medicines are removed; statistics stay correct

Safety, privacy and the rules the app enforces

PATIENT SAFETY

- "Not a diagnosis" reminders on AI, symptom checker, reports and analytics
- Emergency red flags (chest pain, difficulty breathing, bleeding, stroke signs) point to Edhi 115 and Rescue 1122
- No invented numbers: empty states instead of fake charts
- Doctor-written information is clearly separate from automated summaries

DATA PROTECTION

- Passwords hashed (bcrypt); signed tokens with expiry
- Server-side role checks on every API
- A patient can only read their own data; a doctor only their own patients
- Admin screens manage accounts and platform health, not clinical records

VALIDATION EXAMPLES

Situation	Result
Double booking of one slot	Refused (409)
Booking a past or impossible date	Refused (400)
Weight 9000 kg or BP 999/999	Refused (400)
Doctor opens another doctor's patient	Not found (404)
Patient calls an admin API	Forbidden (403)
Forged or expired token	Unauthorised (401)
Weak password / duplicate e-mail	Refused with a clear message

Tested: 49 end-to-end checks across the three roles in the final round, plus earlier suites for admin, doctor (140 checks) and booking (29 checks); every page loaded without console errors on desktop and phone width.

Market research 1/3: what already exists (global)

Product	What it does	How we differ
Ada Health	AI symptom assessment and care guidance; huge user base; reviewed by doctors	No Urdu, no Pakistani hospitals or doctor booking
Infermedica / Symptomate	Medical-grade symptom checker API, 24 languages	Sold to enterprises; no Urdu; no local doctors
K Health	AI chat that connects to doctors in the US; raised \$50M in 2024	US-only, insurance-based
Healthily, Buoy	Self-care and symptom checking apps	General Western content
Babylon Health	AI + video GP (UK / Rwanda)	Collapsed into bankruptcy in 2023 — a warning about over-promising
ChatGPT Health, Microsoft Copilot Health (2026)	Large-model health assistants for personal records (US first)	Not local, not integrated with local doctors, not for Pakistan yet

Lesson: global tools are strong at triage but weak locally. The big failure (Babylon) came from promising too much, so our "assistant, not doctor" positioning is deliberate.

Market research 2/3: what exists in Pakistan

Product	What it does	Gap we address
Marham	Doctor search and booking; claims 14,000+ doctors and 10M+ users	Directory first; limited AI explanation of reports
Oladoc	Booking, online consultation; claims 25,000+ doctors; "Fit by oladoc" wellness AI	Marketplace focus; no full doctor-side screening workspace
Sehat Kahani	Telemedicine run by women doctors; community e-clinics	Telehealth, not an AI health companion
Dawaai	Online pharmacy and lab tests	Pharmacy / diagnostics, not consultation
Ilaaj AI (closest AI rival)	AI health chat in Urdu / Roman Urdu with voice and doctor review, priced PKR 200–1,500	Urdu and voice: our biggest gap to close
Hospital WhatsApp bots (AKUH 2023, Shifa)	Appointment and information bots for one hospital	Single-hospital; we aim to span providers

Honest take: Pakistan already has strong booking and telehealth players. We should not try to out-list them; we win by being the *understanding and tracking* layer (AI explanations + a complete doctor and admin workspace) and by partnering with them and with hospitals.

Market research 3/3: size and signals

USD 946 B

global digital health by 2030 (Grand View Research)

USD 1.4 B → 4.3 B

symptom-checker chatbots 2025 → 2030 (indicative)

~USD 546 M

Pakistan digital health by 2028 (Statista, indicative)

194 M

mobile connections in Pakistan

TAILWINDS

- Young, mobile-first population; internet use growing every year
- Doctor shortage and high out-of-pocket costs push people toward cheaper first advice
- Hospitals already invest in WhatsApp and AI bots, so partners exist
- Large-language-model quality and cost keep improving

HEADWINDS

- Trust: studies (Oxford / Nature Medicine, 2026) show chatbots can mislead real users
- No enacted data-protection law yet (draft bill), so we must self-regulate
- Telemedicine licensing (DRAP / PMDC) and clinical liability
- Language: most people prefer Urdu / Roman Urdu

Market-size figures come from commercial research firms and vary by method. Treat them as indicative, not exact.

What makes Medisynix different

1 · One workspace for all three roles

Most apps serve patients *or* doctors. We connect patient, doctor and administrator on one data model, so what one does is instantly visible to the other.

2 · Explain, don't just answer

Reports come with a plain-language summary, a confidence score and the reasoning steps behind it. The goal: understandable and checkable AI.

3 · Local grounding

The assistant is grounded on Pakistani hospital information (Aga Khan University Hospital, Al Shifa) and prices in rupees, not on generic Western content.

4 · Safety by design

Positioned as information, never diagnosis. Red-flag escalation, disclaimers, and doctors in the loop.

5 · Works even offline from a database

Automatic fallback to local files means a clinic with poor infrastructure still gets a working system.

Where we are not (yet) different

Urdu and voice (Ilaaj AI has it), the size of doctor networks (Marham, Oladoc), and clinical validation. These are our next priorities.

Pros and cons

Strengths

- Complete working product: 3 roles, ~40 pages, validated APIs
- Real AI chat, transparent rule-based tools
- Clean, modern, responsive interface (desktop and phone)
- Strong server-side rules and tested edge cases
- Runs without a database; easy to deploy
- Honest design: clear "not a diagnosis" messages, no fake data
- Low cost to run; open code on GitHub

Weaknesses

- Report explainer is a prototype; symptom checker covers only 10 conditions
- AI chat depends on a third-party service (Chatbase) and internet
- No Urdu / voice yet; no payments, video calls or chat with doctors
- Not clinically validated; no regulatory approval
- Attachments stored with records (2 MB), not in cloud storage
- Notification read-state is per browser
- Small seed dataset; no real hospital integration yet

Opportunities

- Urdu + voice assistant for first-time smartphone users
- Partner hospitals, labs, pharmacies and insurers
- Corporate wellness and community health workers

Threats

- Well-funded local players adding AI (Marham, Oladoc, Ilaaj)
- Trust and liability if the AI is wrong
- Evolving data-protection and telemedicine regulation

Known limitations (say them before someone asks)

Limitation	Why it exists	How we fix it
Report explanation is a sample	Time and cost of integrating a medical model	Connect the uploaded file to a vision / language model with doctor review
Google / GitHub sign-in need keys	OAuth credentials are not in the repo	Add client IDs in .env.local on deployment
MongoDB not running on the demo laptop	Local development uses file storage	Use MongoDB Atlas in production (code paths exist)
Chat depends on Chatbase	Fast way to ship a real LLM with a knowledge base	Own retrieval + model with logging, cost control and evaluation
No clinical validation	FYP scope	Pilot with a clinic, measure accuracy and safety with doctors
No Urdu / voice	Content and speech model work	Roman Urdu and Urdu first; voice via speech-to-text
Data protection law not enacted	Pakistan's bill is still a draft	Follow HIPAA / GDPR-style consent, encryption and audit logs now

How we make it better: roadmap

Next 3 months

- Roman Urdu / Urdu interface and chat
- Connect report upload to a medical model (with doctor review)
- Expand symptom checker; red-flag escalation to a nurse line
- Privacy centre: consent, export, delete my data
- Evaluation set: test the AI on 200 reviewed cases

3 – 9 months

- WhatsApp channel for low-bandwidth users
- Lab and hospital integrations (HL7 / FHIR)
- e-prescriptions and medicine reminders
- Video consultation and secure chat with doctors
- Payments (JazzCash / Easypaisa / cards)

9 – 18 months

- Clinical pilot with a partner hospital and published results
- Insurance and corporate-wellness plans
- Chronic-care programmes (hypertension, diabetes)
- Regulatory approvals and ISO 27001-style audit
- Wearable data (blood pressure, glucose, steps)

How people will reach Medisynix

Meet people where they are

WhatsApp and a lightweight, mobile-friendly web app in Urdu; short videos and health tips on social media.

Partner for trust

Hospitals, clinics, labs and pharmacies recommend it; doctors review content. Universities and medical colleges run pilots.

Reach underserved areas

Community health workers and Lady Health Workers use it to explain results and refer patients; kiosks in pharmacies.

Business model

Free for patients' basic use. Revenue from clinic subscriptions, hospital integrations, corporate wellness and insurer partnerships.

Grow responsibly

Start with one city and two partners, publish accuracy results, then scale. Honesty about limits is part of the brand.

Success measures

Users who understood their report · visits booked · time saved per doctor · red-flag cases correctly escalated · patient satisfaction.

How it can make lives better

The worried parent

At 2 a.m. a child has fever. The symptom checker explains likely causes, what to do tonight and when to rush to hospital, in clear words, instead of panic and random web searches.

The patient with high blood pressure

She records her readings, sees colour-coded trends and books a cardiologist. The doctor sees her history before she walks in, so the visit is about treatment, not paperwork.

The overloaded doctor

AI analysis shows who needs attention first. Notes are written once and appear to the patient. Less admin, more care.

The small clinic

An administrator can set up staff accounts in minutes, export reports and keep the system secure, without an IT team.

The first-time smartphone user

With Urdu voice (roadmap) they can ask a question out loud and get an answer they understand.

The big idea

Better understanding → earlier care → fewer emergencies and lower costs for families and for the health system.

Likely questions and good answers (1/2)

Is the AI real?

The chat is a real language model (Chatbase) grounded on AKUH and AI Shifa information. The symptom checker and vitals screening are rules we can explain. The report explainer is a prototype.

Does it replace a doctor?

No. It gives information and helps people reach the right doctor. Every screen carries a "not a diagnosis" reminder.

How accurate is it?

We have not clinically validated it yet; that is the first step of the roadmap (evaluation set + pilot with doctors).

Where is the data stored and is it safe?

MongoDB, or local files if the database is down. Passwords are hashed, tokens expire, every API checks the role. For production we would add encryption at rest, audit logs and consent.

What if the AI gives wrong advice?

Answers are general, include when to see a doctor, and emergency phrases escalate to Edhi 115 / Rescue 1122. We would log and review chats in a pilot.

Why Chatbase?

It let us ship a working knowledge-based assistant quickly. Later we can swap in our own retrieval + model.

Likely questions and good answers (2/2)

Marham and Oladoc already exist — why this?

They are great at finding doctors. We focus on understanding and tracking health with AI, and on giving doctors and admins a complete workspace. We would partner with them, not fight them.

How will you make money?

Basic use is free. Revenue comes from clinic subscriptions, hospital and lab integrations, corporate wellness and insurer partnerships.

Can it scale?

The Next.js app is stateless; with MongoDB Atlas it can be deployed on Vercel or any cloud and scaled horizontally. AI cost is per message and can be limited.

What was the hardest part?

Keeping three roles consistent: booking rules, permissions and the same data shown correctly to patient, doctor and admin, with the system staying safe when things fail.

What would you do with more time?

Urdu and voice, a real report-reading model with doctor review, payments and video, and a clinical pilot.

Is it legal?

It is positioned as information, not diagnosis or treatment. Pakistan's data-protection bill is still draft, so we follow good-practice consent and security now and would seek DRAP / PMDC guidance before offering consultations commercially.

Demo-day checklist and "if something goes wrong"

MORNING OF THE DEMO

- Charge laptop, plug in power, disable sleep and notifications
- **npm run demo:reset** (fresh data dated for today)
- **npm run build** then **npm start**; open localhost:3000
- Log in once as each role in its own browser window; leave them open
- Test internet: open the General AI Doctor and ask one question
- Zoom 100%, full screen; close other apps
- Keep this PDF open on a second screen or phone

IF SOMETHING GOES WRONG

Problem	Do this
Page shows an error	Press F5. If it persists, restart npm start
AI chat is blank or slow	Check internet; click <i>Start New Chat</i> ; or show the screenshot slide
'Invalid credentials'	Choose the right role tab on the login page
Doctor has no visits today	Run npm run demo:reset and refresh
Port 3000 busy	Close the old terminal or run npm start -- -p 3001
Data looks messy after rehearsal	npm run demo:reset

Golden rule: demonstrate the happy path; mention limits yourself (slide 7) before anyone else does.

Sources and caveats

- DataReportal, *Digital 2026: Pakistan* — population 256 M, internet users 117 M (45.6%), mobile connections 194 M.
- Government of Pakistan, *Economic Survey 2024-25* — about 319,000 registered doctors; health spending under 1% of GDP.
- Pakistan National Health Accounts 2019-20 — out-of-pocket about 52.8% of total health expenditure.
- Grand View Research — global digital health about USD 946 B by 2030. Statista — Pakistan digital health about USD 546 M by 2028 (base year not verified). Market-size sources for symptom-checker chatbots are indicative.
- Competitor information from company websites and press, searched 4 Oct 2026: Ada Health, Infermedica, K Health (funding July 2024), Babylon (US Chapter 7, Aug 2023), Healthily, ChatGPT Health (7 Jan 2026), Microsoft Copilot Health (12 Mar 2026), Marham, Oladoc, Sehat Kahani, Dawaai, Ilaaj AI, Aga Khan University Hospital WhatsApp assistant (July 2023), Shifa International WhatsApp service.
- Oxford / *Nature Medicine* (Feb 2026) on the limits of LLM health advice for lay users.

Caveat: company claims such as "25,000 doctors" are self-reported. Prices and features change; double-check any number you plan to quote aloud. Screenshots in this deck were taken from the running application with the demo data.



Thank you

Questions? Let's talk about making health information understandable for everyone.

github.com/dvlprasher5

Medisynix · AI-powered medical assistant